CROSS-CATCHMENT COLLECTIVE

"From the Catchment to the Village"

Pathways for Delivering Water for Basic Human Needs using the instrument of the BHNR in rural municipalities

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By the

Rivers of the Lowveld Collective

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Full Title

Integrating the Basic Human Needs Reserve into Catchment-to-Village Water Planning **in rural areas of South Africa**

Collaborating Partners

Association for Water and Rural Development (AWARD)

Mahlathini Development Foundation (MDF)

In collaboration with water management institutions, local government and communities in the catchments of the Lowveld.

Rivers of the Lowveld Collective

Abbreviations

Table 1: Abbreviations and acronyms used in this report

Acronym	Meaning
BHNR	*Basic Human Needs Reserve*
ER	*Ecological Reserve*
CMA	*Catchment Management Agency*
DWS	*Department of Water and Sanitation*
WSA	*Water Services Authority*
WSP	*Water Safety Plan / Water Service Provider (context-specific)*
RIDe	*Resources, Infrastructure, Demands and entitlements*
MUS	*Multiple-Use Services*
SWELL	*Participatory planning approach for water services and livelihoods*
MTTR	*Mean time to repair*
MEL	*Monitoring, Evaluation and Learning*
SLA	*Service-level agreement*
VWISA	*Village Water Infrastructure Savings Association (community micro-fund mechanism)*
C2	*Category C2 district municipality (rural district WSA)*
NRW	*Non-revenue water*

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Preamble

Introduction

This exploration is conducted under the Cross-Catchment Collaboration (CCC) funded by the Lewis Foundation. The aim is to bring focused attention to the critical issue of water allocation through the lens of the Basic Human Needs Reserve (BHNR). The exploration is being undertaken jointly by two CCC members, the Rivers of the Lowveld Collective (RLC) and the Northern Drakensburg Collaboration (NDC).

This draft report proposes the development of a scalable, rights-based framework to integrate the Basic Human Needs Reserve (BHNR) for "*catchment-to-village*" water planning and delivery systems. The intention is to generate a working model that can be shared across catchments, ensuring that this critical dimension of water allocation receives consistent attention nationwide.

Background

South Africa has made significant legal and policy advances in recognising the right to water, including embedding it in the Bill of Rights and enacting enabling legislation such as the National Water Act and the Water Services Act. Yet, the gap between constitutional promise and lived reality remains wide. Rural communities continue to experience persistent water insecurity due to structural, institutional, and practical challenges.

The BHNR, enshrined in the National Water Act (1998), is a legally mandated allocation designed to secure every person's right to sufficient water for drinking, food preparation, and personal hygiene. As part of the broader Reserve—which also includes the Ecological Reserve—it enjoys the highest priority in water allocation decisions. The Free Basic Water (FBW) policy, introduced in 2001 to operationalise this mandate, provides 6,000 litres of water per household per month. While this policy framework is robust, practical implementation has been uneven. Many rural communities lack the infrastructure to access their allocation, monitoring and enforcement mechanisms remain unclear, and the benchmark of 25 litres per person per day often fails to reflect the full range of needs in rural contexts, particularly where multiple-use water supports small-scale production and livelihoods.

Challenges and Criticisms

Despite its intentions, the implementation of the Basic Human Needs Reserve faces several challenges:

- Insufficient allocation: Critics argue that the 25 liters per person per day may not be adequate to meet all basic needs, especially in larger households or areas with higher water demands.
- Infrastructure limitations: Some communities, particularly in rural areas, lack the necessary infrastructure to access the allocated water, rendering the policy ineffective in those regions.
- Monitoring and enforcement: There is ambiguity regarding which governmental body is responsible for monitoring and enforcing the Reserve, leading to potential lapses in ensuring that the allocated water reaches those in need.
- **Rural needs:** Despite its legal standing under the National Water Act, the 25 litres per person per day allocation does not reflect the full range of needs in rural contexts, especially where multiple-use water is vital for small-scale production and local income generation.

Purpose of this case study

There is currently **no integrated framework or set of pathways** to guide how municipalities, Catchment Management Agencies (CMAs) and communities should plan, allocate, and account for the BHNR—nor how to uphold accountability through the international **human rights framework**.

This study proposes the development of a **scalable, rights-based framework** that relies on a number of pathways that integrate the **Basic Human Needs Reserve (BHNR)** into **catchment-level** water planning and delivery systems. The framework is **co-developed in partnership at three levels: national, regional/catchment/district and with rural villages—in the Lowveld Rivers region**, serving as a pilot for wider application in South Africa's rural catchments through the CCC.

Rationale and problem statements

- BHNR is a legal priority, but many rural water supply systems are unreliable (downtime, poor quality, access distances greater than 200 m).
- Bulk schemes alone are fragile; governance and O&M gaps undermine functionality.
- Households practice MUS already; planning and tariffs seldom reflect this, causing inequities and hidden costs (e.g., purchasing water from tankers).

Opportunity: integrate catchment allocation (Reserve first), municipal service provision and community-level reliability in one plan, leveraging solar boreholes, standardised spares, service SLAs and community finance (e.g., VWISAs).

Theory of change

If (a) BHNR is guaranteed in allocation rules; (b) infrastructure is standardised and maintained under performance-based O&M; (c) communities co-manage day-to-day tasks with transparent micro-finance; and (d) MUS is recognised and priced post-BHNR, then households reliably obtain >=25 L/person/day close to home, water quality is safe, time and health costs decline, and modest livelihood uses grow without compromising the Reserve.

Figure 1 The BHNR is currently taken as 25 litres per person per day under the RDP standard

Assumptions and dependencies

1. Adequate dry-season yield and/or viable augmentation options for priority supply nodes.
2. Enforceable SLAs with clear penalties and incentives.
3. Visible tracking and routine performance reporting.
4. Tariff approach is accepted and indigent households are protected.
5. Security measures for high-value assets (e.g., pumps, panels).

Key Objectives

- **Map current practices**: Understand how the BHNR is currently addressed (or not) in water service delivery and planning at village, municipal and catchment levels.
- **Co-develop municipal and village-level practices**: Work with local government and communities to **identify and scope planning practices** that enable BHNR realisation in rural communities.
- **Explore catchment integration**: Identify how BHNR delivery at village level can be **linked to catchment-level processes**, including engagement with CMAs and DWS.
- **Introduce rights-based accountability**: Align the BHNR framework with **human rights principles**, building on AWARD's guidance for rights-based water governance.
- **Review BHNR thresholds**: Examine the **suitability of the 25 litres/day benchmark** in supporting basic needs, productive use and rural livelihoods.
- **Lay the groundwork for broader adoption**: Produce **action pathways** that can be adapted and scaled across other local municipalities and catchments.

Approach and Methodology

The opportunity exists to integrate **catchment allocation** ('Reserve first'), **municipal service provision**, and **community-level reliability** in one plan, leveraging **practices under a number of Action Pathways.** The project will elucidate pathways that connect **national policy** with **village-level realities**. This ensures that rights enshrined in law are translated into practice, and that accountability runs through all levels of governance. Activities are listed under each of the proposed pathways.

1. National Pathways: Policy and Legislation

   - Review how the BHNR is framed within the National Water Act, Water Services Act, and Free Basic Water policy.
   - Assess national guidance on the Reserve (e.g., thresholds, monitoring, and reporting requirements).
   - Identify how a human rights framework — defining obligations and violations – can be applied in a real context.
   - Establish the baseline against which catchment and local practices can be evaluated.

2. Catchment Pathways: Regional Water Resource Management

   - Analyse how Catchment Management Agencies (CMAs) and Catchment Management Strategies (CMSs) interpret and implement the Reserve, including the BHNR.
   - Examine allocation processes to ensure that BHNR requirements are prioritised before other uses (agriculture, industry, etc.).
   - Explore how CMAs monitor compliance and respond to violations of the Reserve.
   - Test mechanisms for linking village-level data and needs to catchment decision-making.

3. Services delivery Pathways: Municipal and Water Services Authorities

   - Review how Water Services Authorities (WSAs) and municipalities plan for and deliver the BHNR through Integrated Development Plans (IDPs) and Water Services Development Plans (WSDPs).
   - Assess alignment between municipal service delivery (e.g., Free Basic Water provision) and catchment-level allocation decisions.
   - Identify institutional overlaps or accountability gaps that undermine Reserve delivery.
   - Support municipalities to integrate rights-based accountability tools (monitoring, reporting, grievance mechanisms).

4. Village Pathways: Community and Household Practice

   - Partner with pilot villages (under various current projects of AWARD, MDF and K2C) in the Lowveld Rivers region to reflect on practical approaches to realising the BHNR.

- Work with village water committees, households, and local leaders to:
- Track water quantity, quality, and accessibility.
- Identify gaps between entitlements and actual delivery.
- Identify simple tools for community monitoring and reporting.

The Basic Human Needs Reserve [BHNR]: The Basics

According to South Africa's legal framework access to water for basic needs water is a constitutional, justiciable socio-economic right, and then gives that right operational meaning through the "Reserve" in water law. (du Toit, Pollard, & Pejan, n.d.). We start with an overview of the legal framework, the basic concept and how the BHNR can be calculated for delivery.

Figure 2 People queuing at a community standpipe in Bushbuckridge

Legal framework: water as a basic human right

Constitutional foundation: the right to sufficient water and a protected environment

The Constitution places the right of access to "sufficient water" in the Bill of Rights and, importantly, South Africa's Constitutional Court has affirmed that socio-economic rights are justiciable—meaning they can be evaluated by courts and create enforceable duties for the state. (du Toit, Pollard, & Pejan, n.d.)

The document also emphasises that the constitutional right to water is connected to environmental rights (Section 24), because the long-term ability to realise water rights depends on sustaining aquatic ecosystems that produce water of adequate quality and quantity. (du Toit, Pollard, & Pejan, n.d.)

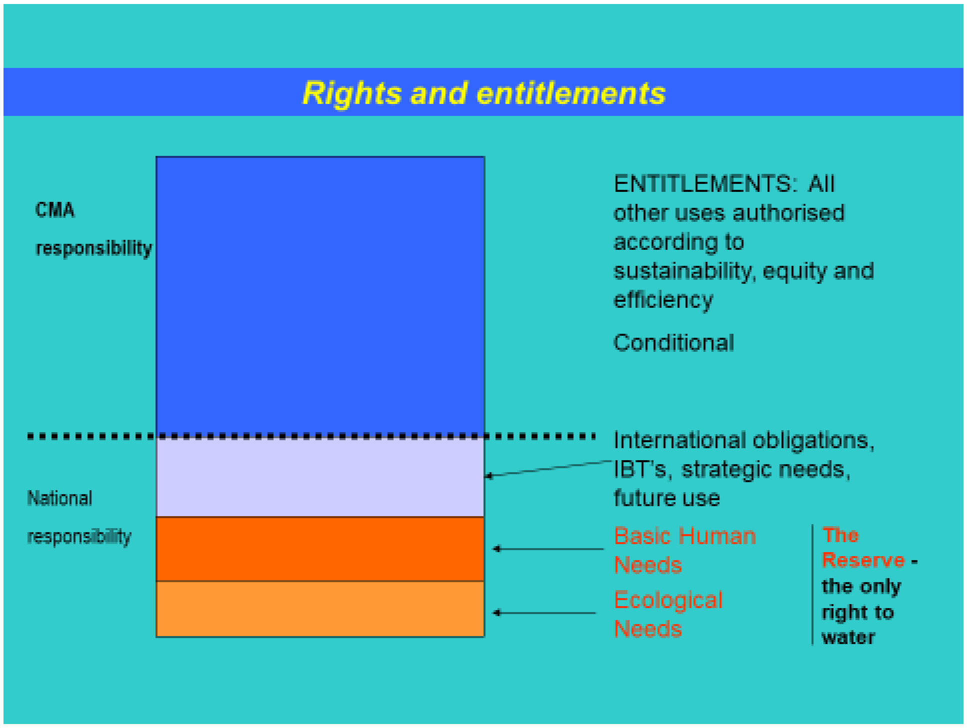


Figure 3 The Reserve is the only right to water and has the highest allocation priority. (in red)

Translating rights into water law: the National Water Act and the "Reserve"

While the Constitution frames the right, the National Water Act (NWA) is described as the instrument that gives it specific legal meaning, supported by national strategy and catchment strategies that concretise policies and standards. (du Toit, Pollard, & Pejan, n.d.)

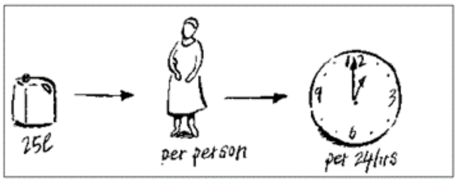
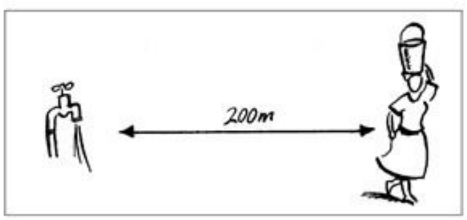
Under the national policy principles underpinning the NWA, the document stresses a crucial legal shift: there is no ownership of water, only (i) a right for basic human needs and the environment and (ii) authorisations for other uses. Within this framework, the Reserve is defined as the water for basic human needs (BHNR) and ecological needs (ER) and is given priority "by right", while all other uses are subject to authorisation. (du Toit, Pollard, & Pejan, n.d.)

What BHNR is (in law and policy) and its relevance for water allocation planning

In this rights-based framing, the document notes that the BHNR has been quantified in operational terms as a minimum of 25 litres per person per day available within 200 metres (a practical expression of "accessibility" and "minimum content"). (du Toit, Pollard, & Pejan, n.d.)

It also makes explicit the hierarchy this creates for allocation: the Reserve is deducted first from the available resource before water can be allocated to other uses; thereafter, remaining water may be licensed/authorised for sectors like agriculture, mining and industry, but licences cannot be issued in a way that ignores the Reserve without violating constitutional obligations. (du Toit, Pollard, & Pejan, n.d.)

Table 2: Components and quantification of the Basic Human Needs Reserve

BHNR component	Details
	The figure that South Africa has currently set for basic human needs is 25 litres of water per person per day (the RDP minimum) for basic needs at the household level.
	Provided within a 24 hr cycle
	Of adequate quality (SANS 241 standards)
	Within 200m of households

Duties on the state: respect, protect, fulfil (and progressive realisation)

The document uses the human-rights "tripartite" duty framework to explain what the right to water requires of government:

- Respect (do not interfere with enjoyment of the right);
- Protect (prevent third parties from undermining the right); and
- Fulfil (take positive steps—planning, resources, capacity, systems). (du Toit, Pollard, & Pejan, n.d.). "Fulfil" is linked to the practical requirements of implementation—especially monitoring capacity and systems (e.g., data, infrastructure, institutions) needed to ensure the Reserve is actually maintained. (du Toit, Pollard, & Pejan, n.d.)

The internal report also notes that socio-economic rights are commonly evaluated using a reasonableness standard and through progressive realisation—requiring measures that are deliberate, concrete, and not retrogressive, moving "as expeditiously as possible" toward full realisation. (du Toit, Pollard, & Pejan, n.d.)

Table 3: Defining Obligations under a human rights framework

Obligations		
Negative	Positive	
Respect	*Protect*	*Fulfil*
The State shall NOT…	The State SHALL…	
(Examples only)	(Examples only)	(Examples only)
• Support actions that negatively affect people's ability to access basic water	• Enact legislation against water pollution	• Facilitate the participatory development of a national water strategy
• Increase one groups access to basic water at the expense of another group's access to basic water	• Fight crime to ensure the safety of people accessing water	• Ensure that information needed for good water management and use is easily accessible
• Develop legislation that works against people's water rights	• Ensure that water plans and practices are ecologically and socially sustainable	• Support, encourage and facilitate the improvement of rural water infrastructure

Accountability architecture: standards, monitoring, and the "public trust" idea

A key point for BHNR reporting is the argument that a rights approach pushes institutions to create definitive, justiciable standards for the Reserve, because without enforceable standards and monitoring, the Reserve risks remaining a policy concept rather than something that can be demonstrated, challenged, or corrected. (du Toit, Pollard, & Pejan, n.d.)

It further links accountability to the idea of water as a public trust, with the Minister as trustee, implying a duty to ensure decisions do not violate the public's right to water; the paper notes how the precautionary principle can shift burdens toward the state to show that combined developments and licences will not result in violations (especially in relation to ecosystem protection). (du Toit, Pollard, & Pejan, n.d.)

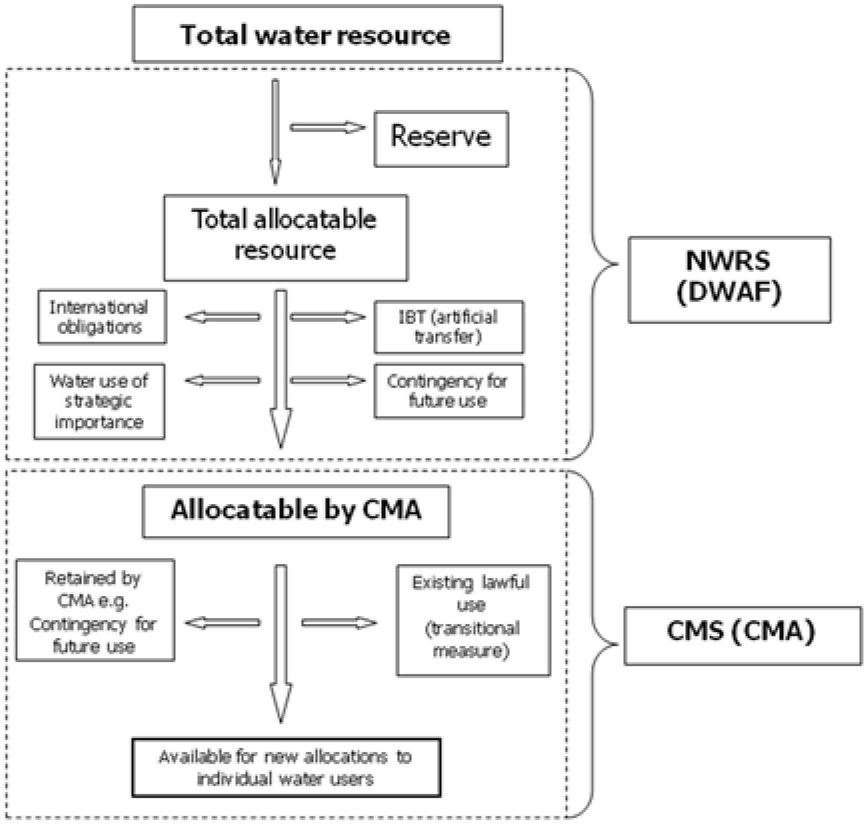


Figure 4 Overall water allocation responsibilities. The Reserve is the only right to water and has the highest allocation priority. Existing lawful use gives temporary entitlements until all users are licensed through compulsory licensing (IBT = interbasin transfer)

Providing the BHNR: Application of the RIDe lens

The frustration around Basic Human Needs Reserve (BHNR) implementation is not that the idea is unclear, but that it is too often treated as a small, technical number in a catchment water balance, rather than a priority obligation that must translate into reliable water at the point of use. The ability to realise BHNR depends on how Resources, Infrastructure, Demand and Entitlements (RIDe) (see later) interact—over time, under uncertainty, and under political pressure. From a RIDe perspective, this mismatch is predictable: resources are only one part of the story.

From a RIDe viewpoint, the BHNR “implementation gap” isn’t a technical glitch—it’s a systems problem. Resources without infrastructure deliverability, demand without credible projections, and entitlements without ring-fencing and proof will keep producing the same outcomes: small numbers, rounding-to-zero optics, rising conflict, and households still exposed to intermittent, unreliable access.

Notes from the field

There seems to be a lack of information regarding how the basic human needs reserve have been actioned subsequent to the developing of the policy. The new norms and standards for water services provision suggested by Mark Bannister (DWS chief engineer) should hold adequate information regarding the calculations and the provision for the reserve for the basic human needs reserve, but that is not clear.

There are problems around some of the statistics to calculate on the basis of the latest figures and then escalate for a further 10 or 20 to ***25 year*** *projection in terms of population growth. Some issues have come up where in the water balance, the figures for the basic human needs reserve provision are so small compared to the big users like agriculture and mining and so on, that when the figures get entered into any table, they round up to or round down to a zero, which doesn't provide very convincing data because it shows in these tables as a zero. And of course, there's a bunch of people concerned that zeros on the basis of opportunity catchment sees is a completely false reflection on what should be allocated to that reserve for basic services and basic human needs provision.*

Jackie Jay worked on the ecological reserve and voiced concerns on the BHNR implementation. If the BHNR is escalated in terms of demand from population or if the allocation is stepped up from 25 liters per person per day to, say, 100 or 110, the water balances are either pushed into deficit or, pushed further into deficit. What that would likely precipitate is the re-allocation of water for basic human ***needs from*** *the ecological reserve or licensed users. What's likely to happen is all that there will be resistance from agricultural and industry sectors, a process going to meet with resistance.*

Opposition argues that if allocations to the municipalities are stepped up and the basic human needs reserves escalated through increased volumes, much of that water could be lost through non-revenue water and leakages. Also the argument is that the additional provision could end up allocated, through municipal supply, to light industry and commercial through municipal licences.

This generally presents a big risk for provisioning for the basic human needs reserves and will probably continue to be a massive challenge unless methodologically more clarity is provided and also that clear guidelines and ring-fencing of the BHNR are tabled and so that constitutional right to water can be met without encroaching upon the ecological reserve. And these are some of the important issues that have arisen ***in conversations****.*

Worth noting is that the CMAs are not responsible for actual provision of the basic human needs reserves, but simply allocating it from the total volumetric availability through the mean annual runoff calculation. These are the big issues that need to be factored into the way for catchments meeting their constitutional obligation to provide water for basic human needs reserves and free basic water.

Currently there is "no integrated framework" that clearly shows how municipalities, CMAs and communities should plan, allocate, and account for the BHNR, and monitoring/enforcement responsibilities remain ambiguous in practice.

*What those consulted raise is exactly where BHNR planning tends to break: it's treated as a small line item in a catchment water balance, rather than a priority obligation that must be translated into deliverable se****rvices, with losses****,* ***governance*** *and safeguards made explicit.*

The RIDe methodology and application in this BHNR report

Work by the IRC and AWARD (2006) makes clear that BHNR challenges are not limited to "how much water exists in the catchment". Rather, they arise from interacting constraints across the full chain of delivery. The RIDe framing (Resources, Infrastructure, Demand and Entitlements) helps keep the whole chain visible:

Resources: Is sufficient water available at the right time and place after protecting the Reserve?

Infrastructure: Can allocated water be abstracted, treated, conveyed and distributed reliably to households?

Demand: How will population growth, settlement change, service-level targets and economic uses evolve over 10-25 years?

Entitlements: Are priorities clear, measurable, ring-fenced and enforceable, and can people access remedy when they are not met?

A BHNR plan that addresses only Resources will repeatedly fail on the other three dimensions, producing "paper compliance" and continued household insecurity

This report uses the **RIDe framework—Resources, Infrastructure, Demand and Entitlements**—as a practical method for analysing why the **Basic Human Needs Reserve (BHNR)** often remains visible in policy and catchment water balances, yet weakly realised as **reliable water access at household level**.RIDe is used because BHNR outcomes are rarely explained by a single factor. In many catchments the BHNR "exists" hydrologically (or numerically), but households still experience intermittent supply, long walking distances, poor water quality, broken pumps, or systems that prioritise other uses in practice. RIDe makes these interacting causes visible by forcing analysis across the whole delivery chain—from water availability to operational performance, and from legal protections to the lived experience of access.

What RIDe covers

1) **Resources**. This dimension asks: *Is water available in the system, in the right place, at the right time, and of adequate quality?* It includes groundwater and surface water availability, seasonal variability and drought risk, water quality constraints, and the practical "exploitable" portion of a resource (not only theoretical yield). For BHNR, this avoids overconfidence in paper volumes and highlights whether minimum needs can be met under dry-year conditions without undermining the ecological reserve.

Figure 5 Water resources availability in the Lowveld are frequently subjected to climatic conditions

2) **Infrastructure.** This dimension asks: *Can available water be converted into delivered services?* It covers bulk conveyance, treatment capacity, distribution networks, boreholes, pumps, storage, and operational functionality. In rural contexts, infrastructure performance is often the binding constraint. RIDe explicitly considers non-revenue water (NRW), breakdown frequency, maintenance systems, and whether assets are fit for context. For BHNR, this is where "allocation at source" either becomes "water at the tap" or disappears into losses, downtime, or inequitable distribution.

Figure 6 What is meant by infrastructure in a rural setting is fluid set of contributions

3) Demand This dimension asks: *Who needs water, how much, for what purposes, and how is demand changing over time?* It includes domestic demand for minimum needs, household productive uses, institutional and commercial demand within municipal systems, and major catchment users such as irrigation, forestry, mining, and industry. It also considers population growth and settlement patterns, and the way latent demand emerges as infrastructure expands. For BHNR, RIDe helps clarify the difference between the protected minimum (the BHNR) and higher service-level targets, and supports scenario-based planning under uncertainty.

Figure 7 Demands in rural areas are specific and frequently foundational to rural livelihoods and income generation

4) Entitlements This dimension asks: *Who has what rights and responsibilities, and how is compliance demonstrated and enforced?* It includes the legal status of the BHNR within the Reserve, the policy commitments of Free Basic Water, licensing and authorisations for other uses, and the institutional division between water resource management and water services delivery. RIDe is particularly useful for identifying accountability gaps—for example, where a CMA/DWS function may "set aside" water for the Reserve, but municipal systems cannot (or do not) ring-fence and deliver the minimum to households. It also focuses attention on monitoring, reporting, grievance mechanisms, and the conditions needed to make BHNR "auditable" rather than aspirational.

How RIDe is applied in this report

RIDe is applied as a structured diagnostic across the BHNR "pathways":

- **National level**: how BHNR is framed in law and policy, and what minimum service standards and accountability mechanisms support delivery.
- **Catchment level**: how Reserve volumes are determined, represented in water balances, and prioritised relative to large users and ecological requirements.
- **Municipal/service level**: how allocations are translated into planning, budgets, infrastructure operations, NRW control, and emergency response.
- **Village/household level**: how people actually access water (quantity, quality, reliability, distance), including coping strategies and local monitoring.

Each case analysis is presented in a way that helps distinguish between:

- **a resource problem** (not enough water, or poor quality),
- **an infrastructure problem** (water exists but cannot be delivered reliably),
- **a demand problem** (unmanaged growth or competing uses), and
- **an entitlement/accountability problem** (minimum rights not protected, measured, or enforced).

RIDe and BHNR decision-making

Using RIDe allows the report to move beyond general statements such as "there is not enough water" or "infrastructure is failing" by producing:

- a transparent explanation of the BHNR implementation gap,
- a defensible basis for BHNR volumes that accounts for losses and reliability,
- practical entry points for action (NRW reduction, O&M strengthening, service-level scenario planning, ring-fencing), and
- a clearer accountability chain linking allocation decisions to verifiable delivery outcomes.

In short, RIDe is used in this BHNR report because it aligns with the real-world challenge: the BHNR is not only a number in a catchment balance—it is a minimum service obligation that must be made deliverable, visible, and enforceable across the full water system.

Case Study: A RIDe Assessment for the Sand River Catchment

This case study applies the RIDe (Resources–Infrastructure–Demand–entitlements) framework to the Sand River Catchment to understand why legally recognised water needs—especially the Basic Human Needs Reserve (BHNR)—are not consistently translated into reliable water services at village level. The Sand River Catchment provides a revealing example of a rural catchment where water scarcity, competing uses, and institutional fragmentation intersect with aging or incomplete infrastructure and uneven management capacity, producing persistent shortfalls in domestic supply and high levels of vulnerability for poor households.

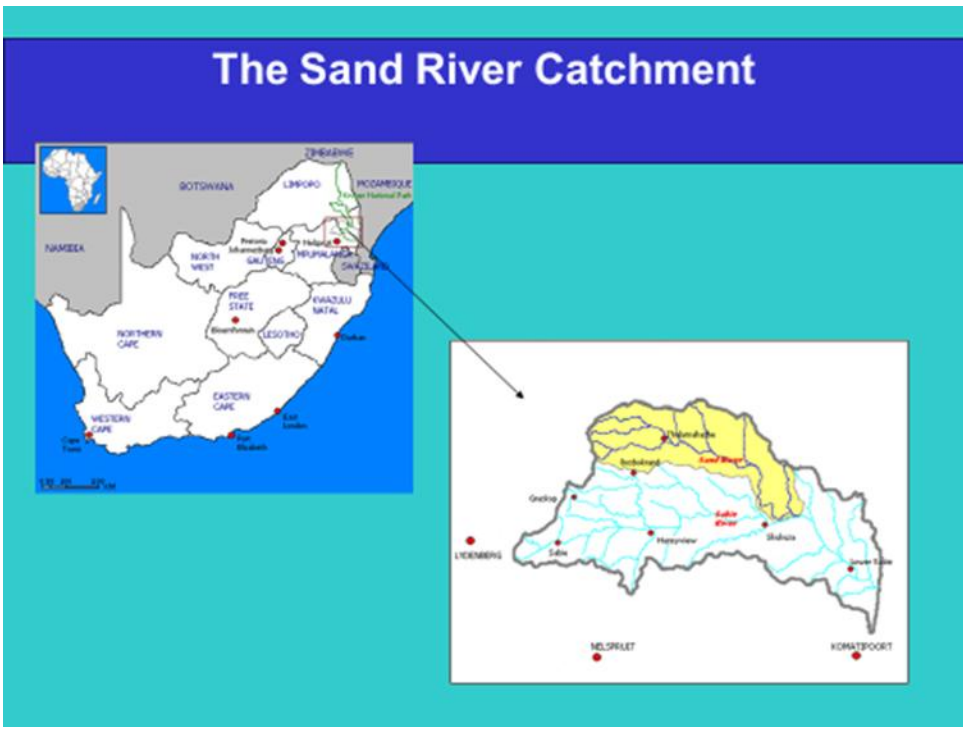


Figure 8 The location of the Sand River catchment in South Africa

In the Sand River case, the value of RIDe is that it makes visible the *"disconnections"* that often sit behind a simple statement like "there is not enough water." These can include: water being available in the resource, but not accessible due to broken infrastructure; infrastructure existing, but not producing services due to O&M gaps, power interruptions, security risks, or poor spares management; and water needs being recognised in policy, but not met because accountability is split across institutions with limited coordination between catchment management and municipal service delivery.

The case study therefore aims to do three things. First, it provides a shared evidence base on the current state of the Sand River system—where water comes from, how it is conveyed, where it fails, and who is affected. Second, it identifies the priority constraints and leverage points for improving BHNR delivery, distinguishing quick operational fixes from longer-term investments and governance reforms. Third, it demonstrates how RIDe can be used as a practical planning method to connect Reserve requirements with service standards, infrastructure rehabilitation, and institutional arrangements that make delivery measurable and enforceable.

The findings from this case study are intended to support a shift from “paper compliance” to operational delivery: a BHNR that is not only calculated and set aside, but is demonstrably delivered as safe, reliable water—especially for the poorest and most vulnerable households in the catchment.

Sand River catchment RIDe Assessment

Resources

Main sources and access patterns

- Domestic supply in the Sand River Catchment (SRC) relies primarily on groundwater (boreholes/springs) and bulk supply from dams, but many households also use direct river abstraction, unprotected springs, hand-dug wells in riverbeds, or private water vendors—showing a mixed, often informal resource base alongside formal schemes (Pollard & Walker, 2000).

Groundwater availability, exploitability, and spatial inequality

- Total groundwater in the SRC is estimated at ~8.8 Mm^3/year, but less than half is considered exploitable, meaning the “paper resource” overstates what can realistically be developed for supply.
- The eastern SRC is especially dependent on groundwater (few alternatives), yet this area’s groundwater is poorer quality than the west/north/south, creating a geographic equity problem where the most dependent areas also face the highest quality constraints (Pollard & Walker, 2000).
- Pollard & Walker (2000) estimated ~60–70% of the population was served by boreholes and springs, confirming groundwater as the dominant resource underpinning rural domestic access.

Drought sensitivity and sustainable yield uncertainties

- In Ward 33 (eastern SRC), Munyai (2008) found that during drought/low rainfall, abstraction increasingly draws on groundwater storage because recharge declines, making supply reliability sensitive to climate variability and pumping intensity.
- Munyai’s assessment suggested the area’s sustainable yield could meet maximum demand, but he also flagged that some villages had limited or insufficient supply from existing developed groundwater, implying that localised hydrogeology and development patterns create supply shortfalls even where regional yield estimates appear adequate.

Groundwater quality risks from sanitation

- A key resource constraint is pollution vulnerability: sanitation systems in Ward 33 were identified as a major threat to groundwater quality due to poor construction and because many production boreholes are located downstream of villages and aligned with groundwater flow, increasing contamination risk (Munyai, 2008).
- This highlights a critical RIDe point: water “availability” is meaningless if quality is compromised, especially where communities rely solely on groundwater.

Surface water infrastructure and limits

- Four “homeland-era” dams exist in the SRC: Acornhoek, Edinburgh, Orinoco, and Casteel.
- Acornhoek and Edinburgh were designed for domestic water supply (DWAF).
- Orinoco and Casteel were developed mainly for irrigation by Dept. of Agriculture, Mpumalanga.
- Edinburgh Dam was intended to support both agriculture and domestic supply in the eastern SRC, but in practice most of its water is pumped into the domestic system (Mallory, 2008; Rowse, 2008).
- Despite these assets, collectively the dams are insufficient to meet resident needs, indicating that built storage does not translate into adequate, reliable supply volumes for the catchment.

Inter-basin transfer and regional dependence

- The SRC is the drier part of the Sabie–Sand system. The Inyaka Dam (and the Inyaka inter-basin transfer) was developed to augment water quantities into the Sand River Catchment (Mallory, 2008).
- This creates a structural dependence on external transfers to stabilise supply, increasing exposure to upstream allocation decisions, operating rules, and inter-catchment competition.

Surface water quality outlook

- Pollard & Walker (2000) described Sand River water quality as generally good, but with elevated turbidity and nutrients at certain points.
- They anticipated that population growth would compromise quality unless management measures—particularly wastewater treatment—were improved. This links resource sustainability directly to settlement growth and sanitation governance.

RIDe “Resources” takeaway for the Sand River catchment

- The SRC has a mixed resource portfolio, but domestic water security is dominated by groundwater, with strong east–west inequities in both reliance and water quality.
- The “headline” availability figures mask three binding constraints:
    1. limited exploitable groundwater,
    2. drought-related recharge/storage sensitivity, and
    3. quality degradation risks from sanitation and rising nutrient loads.
- Surface water storage exists but is inadequate for domestic needs without augmentation, and the inter-basin transfer has become a central stabiliser—introducing external dependency into local basic services planning.

Infrastructure

Infrastructure in the Sand River Catchment (SRC) is a layered mix of:

- Regional/bulk infrastructure (dams, bulk conveyance, treatment works),
- Village reticulation networks (standpipes, pipelines, local distribution), and
- Groundwater abstraction infrastructure (boreholes, pumps, controls). The case shows that many failures are not about “missing assets” but about how inherited, overlapping systems are operated, maintained, and governed over time.

1) Bulk supply network: treatment works and the Inyaka-to-villages backbone

Treatment works as key nodes

- A number of drinking water treatment works (DWTWs) operate across the Sabie-Sand system. Several are operated by BWB (Hoxani, Thulamahashe, Shatale, Dwarsloop, Edinburgh, Inyaka), with Shatale, Dwarsloop, and Edinburgh located in the SRC and forming important SRC supply nodes.

Bulk reticulation: the critical link that has underperformed

- The bulk conveyance system intended to deliver water from Inyaka Dam to villages in the SRC has been under construction for many years.
- Although reported as “largely complete” with many villages connected, supply has remained erratic and dysfunctional (Mallory, 2008). Up to 2008 the system had reportedly not yet functioned as intended.

Construction quality and oversight failures

- A major failure was the reported collapse of sections before commissioning, attributed to sub-standard materials and weak supervision by both the municipality and DWAF regional office (Mr Tibane, pers. comm., 2009, as cited).
- This highlights a central RIDe infrastructure point: asset delivery governance (procurement, quality control, supervision) can be as decisive as hydrology in determining whether basic services are realised.

2) Village reticulation networks: “spaghetti systems,” neglect, and informal connections

Legacy standpipe systems and uneven upgrades

- Many villages have basic public standpipe systems installed by former homeland governments (Gazankulu and Lebowa), and later additions by Bushbuckridge Local Municipality.
- These systems have been badly neglected in maintenance and repair, and have accumulated numerous informal and unplanned private connections (Pollard & Walker, 2000). This undermines pressure, reliability, equity, and water loss control.

Coverage and complexity

- Reticulation covers rural villages and the former R293 towns (Thulamahashe, Dwarsloop, Shatale, Mkhuhlu).
- In some villages, two or three reticulation networks were built on top of each other, reflecting incremental projects over time without a single integrated network design.

Planning assumption that never fully materialised

- Village networks were often built on the assumption they would later be connected to a reliable bulk supply.
- The result is the "spaghetti network" described on the ground in Bushbuckridge (AWARD, 2005): fragmented, overlapping, difficult to manage, and vulnerable to leakage and inequitable distribution.

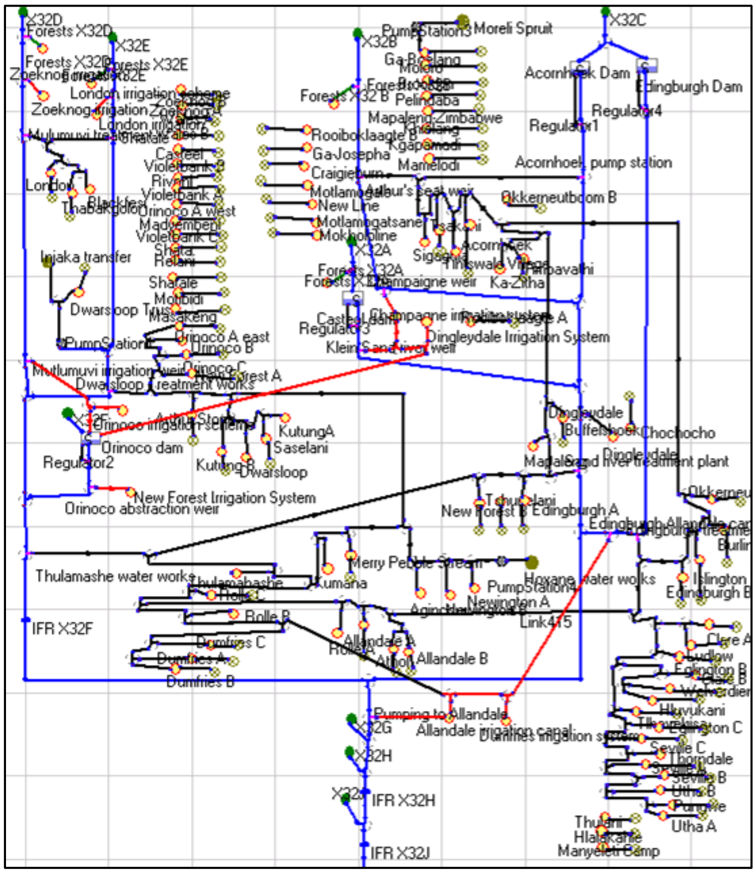


Figure 9 An infrastructure map of the Bushbuckridge supply system showing the complexity and complicatedness in rural systems

Practical implication for BHNR

- Even if bulk water is available, fragmented village networks can:
- fail to deliver minimum volumes consistently,
- amplify NRW and unaccounted-for use,
- privilege households closer to taps or with informal connections,
- and make "ring-fencing" basic supply extremely difficult.

3) Boreholes: dependence, institutional handover, and O&M failure as a service constraint

Boreholes as essential infrastructure

- Many villages depend on groundwater via boreholes; this infrastructure has historically been under DoA or DWAF, later transferred to municipal responsibility when municipalities became WSAs/WSPs.

Institutional complexity and slow response

- DWAF continued operating and maintaining boreholes under its jurisdiction back when it still carried water services responsibilities.
- The cross-border nature of the municipal area reportedly caused long delays in mobilising O&M resources. Reported turnaround times for repair responses could be up to eight months (Mr Tibane & Mr Ndubane, pers. comm., 2005, as cited).

Direct household impact

- When pumps fail and repairs are delayed, broken boreholes remain broken for long periods, forcing villagers to walk long distances to other villages to find water—turning an "infrastructure issue" into an everyday access crisis.

Post-transfer burden and weak capacity

- DWAF was responsible for O&M in most rural villages (with the municipality responsible mainly in the former R293 towns).
- After asset transfer, the municipality became "theoretically" responsible for O&M across rural areas—creating a major added operational burden.

- The overall status of service delivery is described as very poor, illustrating that asset transfer without commensurate capacity, budgets, spares systems, and management controls can worsen rather than improve outcomes.

RIDe "Infrastructure" for the Sand River catchment

The Sand River case demonstrates a compounded infrastructure challenge:

- Bulk systems (Inyaka transfer + treatment + bulk reticulation) exist but have been undermined by construction quality failures and incomplete/erratic functionality.

- Village reticulation is characterised by legacy networks, overlap, neglect, and informal connections, creating high losses and inequitable delivery.

- Borehole infrastructure is essential but has been constrained by slow maintenance response, institutional handover problems, and limited municipal O&M capacity.

The SRC infrastructure story is not simply "insufficient infrastructure"—it is infrastructure that is fragmented, poorly maintained, and weakly governed, which prevents available water (and even allocated BHNR volumes) from being converted into reliable, equitable household supply.

Demand

The requirements of people/ users, and the institutions that represent them

Demand profile: dominance of large licensed/unlicensed productive sectors

- The Sand River Catchment (SRC) demand profile is strongly shaped by productive water use, with irrigation identified as the single largest user (estimated 32.3 Mm^3 in

1985), including commercial plantations (citrus, coffee, mango) and small-scale irrigation (mainly field crops) (Butterworth et al., 2001).

- Forestry/afforestation in the upper catchment (mainly exotic species such as pine) is also a major consumer (estimated 11.3 Mm^3 in 1985), reinforcing that land-use driven demand is structurally significant in the SRC (Butterworth et al., 2001).
- By contrast, domestic water use "to meet minimum needs only" was estimated at ~3.5 Mm^3 (1998)—small relative to irrigation and forestry, which is central to your earlier point that BHNR volumes can look negligible in water balances.

1) Domestic demand: a dual reality (high consumption nodes + widespread unreliable systems)

High-consumption towns

- The towns of Dwarsloop, Thulamahashe, Mkhuhlu and Shatale have fully reticulated systems where consumption exceeds 175 ℓ/c/d (Pollard & Walker, 2000).
- These nodes represent a "higher-service" demand profile within the catchment, likely linked to better reticulation, more continuous supply, and closer alignment with typical municipal consumption norms.

Widespread inequitable and unreliable village supply

- Pollard & Walker (2000) describe the majority of domestic systems in the SRC as providing inequitable and unreliable service, meaning actual household demand is shaped as much by supply constraint as by "need".
- A critical insight is that if you include basic needs plus household productive uses (e.g., small gardens, livestock watering, micro-enterprises), the real domestic demand on local systems may be two to three times higher than "minimum-needs-only" estimates. This anticipates the Multiple-Use Services (MUS) argument: household-level demand is not purely consumptive.

Implication for BHNR planning

- The domestic demand line in many catchment tables is likely an underestimate if it is based only on "minimum needs," because it ignores:
- household productive water use,
- coping strategies (multiple sources),
- and latent demand suppressed by unreliable infrastructure.

2) Environmental demand: a non-negotiable component alongside human needs

- Pollard & Walker (2000) note that environmental demand includes both:
- the Ecological Reserve, and
- the requirement that the river must continue to flow to Mozambique, reflecting international obligations.

- In RIDe terms, the environment functions as a prior claim on flow that must be met before discretionary allocation—so “available water for allocation” is already reduced by these obligations.

3) Demand projections show steep growth pressures and high uncertainty

Sector demand projections (Pollard & Walker, 2000) Their estimates (Table shown in your text) illustrate two major points:

- Irrigation demand growth is projected to be very large (1985: 32.3 Mm^3 → 2010: 70.9 Mm^3).
- Domestic demand is projected to rise dramatically (1985: 3.7 Mm^3 → 2010: 30.8 Mm^3).

These projections underscore that domestic demand becomes politically and hydrologically significant over time, especially as supply networks expand and suppressed demand is released.

Data and monitoring gaps

- A major demand-management challenge is that for key sectors (notably forestry and agriculture), Pollard & Walker (2000) note that consumption figures are not known or monitored, complicating:
- cost estimation,
- verification,
- enforcement,
- and credible water balance modelling.
- They further note that these large users were regarded as “biggest users” and no payment was made for their use at the time, highlighting a governance and incentive gap that affects demand control.

4) Bulk supply volumes and the moving target problem (extended network, changing availability)

Bulk supply as a constraint on realised demand

- Mallory (2008) estimated that bulk supply from Inyaka Dam to the SRC is approximately 10 million m^3/annum, with 2 million m^3/annum transferred onward to major towns in the Sabie catchment.
- Quantifying “the rest of the catchment” demand is difficult because realised demand is dependent on availability, and availability is changing as the Inyaka pipeline is extended to previously unsupplied villages.
- This is a key RIDe demand insight: demand is not static—it expands as infrastructure expands, and it is “revealed” as the system starts to function.

Localised future domestic + productive demand

- Munyai (2008) estimated future domestic and small-scale productive demand in Ward 33 alone for 2020 at 1,210 m^3/day—an example of sub-catchment/village cluster demand that can be material when aggregated, especially if multiplied across wards and adjusted for losses.

RIDe “Demand” for the Sand River catchment

Demand in the SRC is characterised by:

- Dominance of productive sectors (irrigation and forestry) that are large, growing, and historically weakly monitored—creating a major uncertainty and governance challenge in water balance modelling.
- Domestic demand that appears small in “minimum needs” terms but is likely understated when household productive uses and latent demand are included (potentially 2–3× higher).
- Strong spatial differentiation: reticulated towns show high per-capita consumption (>175 ℓ/c/d), while most rural systems remain supply-constrained and inequitable.
- A moving target created by pipeline expansion: as bulk infrastructure reaches new villages, domestic demand increases and becomes harder to quantify using static assumptions.
- Environmental and cross-border flow obligations that function as a prior claim and must be incorporated into any credible demand-and-allocation narrative

Entitlements

In the RIDe framing, entitlements are the “rules of access” that determine:

- who has a right (or a legally protected minimum),
- who holds authorisations/licences for abstraction/use,
- who has the duty to provide services,
- and how accountability is enforced when minimums are not met.

In the Sand River case, the core entitlement problem is not the absence of policy, but the gap between legal entitlement and delivered service—and the way that institutional roles are split across water resource governance and water service delivery.

1) What people are entitled to (the minimum content of access)

BHNR + FBW establish a legal requirement to secure access

- The Basic Human Needs Reserve (BHNR) (as part of the Reserve in the National Water Act) and the Free Basic Water (FBW) policy together create a legal and policy requirement that government must ensure people gain access to water.
- In the “minimum service” framing referenced in your text, the national objective (RDP standard) is:
- 25 litres per person per day (25 ℓ/c/d) of potable water
- supplied from a standpipe within 200 m of each household.

Sand River reality: widespread shortfall against the minimum

- In 2000, AWARD estimated that 44% of the population in the Sand River Catchment were still below the RDP standards for basic water access.

- Community priorities mirrored this: Pollard & Walker (2000) recorded that adequate and reliable provision was a major development need for most communities, later confirmed by other work (Maluleke et al., 2005b).

Entitlement implication

- The entitlement is clear in principle (minimum quantity + accessibility), but the evidence from the catchment suggests that a large share of households were not receiving that minimum—meaning the problem is one of implementation and enforceability, not "policy absence."

2) Who has what right/authorisation (Reserve vs other uses)

Priority "right" vs authorised uses

- BHNR is part of the Reserve and is framed as a priority requirement: it is not a discretionary allocation competing on equal footing with other uses.
- In practice, however, the Sand River demand structure is dominated by large productive uses (irrigation/forestry) and domestic "minimum" needs can appear small in volume terms. This creates an entitlement risk: the minimum right is legally prior but politically and numerically marginalised unless it is explicitly protected in planning and reporting.

The key entitlement tension in the Sand

- Where communities remain below basic standards while large users continue abstracting, the legitimacy question becomes acute:
- Are the priority entitlements (BHNR/basic services) being treated as "first call," or are they being treated as a residual demand that is "met when possible"?
- The field evidence points to the latter outcome in many villages—at least historically—because the lived experience was continuing unreliability and shortfalls against the minimum.

3) Split responsibilities: allocation duty vs service duty

Water resource governance side (CMA/DWS dynamic)

- Under the National Water Act framing, the Reserve (including BHNR) is part of water resource management and allocation. This sits in the governance space associated with the national department (DWS, historically DWAF) and, where established and empowered, Catchment Management Agencies (CMAs).
- In that logic, institutions are responsible for ensuring the Reserve is recognised and protected in catchment allocation decisions.

Water services side (WSA/municipality logic)

- The RDP standard (25 ℓ/c/d within 200 m) and the FBW policy translate the right into a service delivery obligation, which sits with the Water Services Authority/Provider (typically the municipality).

- So the municipality is accountable for ensuring the minimum service is delivered, including the practical aspects: infrastructure, O&M, reliability, and equitable access.

Why the split matters in the Sand River case

- The Sand River case shows the classic “entitlement gap” that emerges when these responsibilities are split:
- A reserve entitlement can exist (in law and in water balances), yet households remain below minimum standards because service systems fail (infrastructure dysfunction, O&M gaps, poor reticulation performance).
- Conversely, a municipality may face pressure to provide services but may be constrained by the resource allocation environment, quality constraints, and bulk system performance.

4) Accountability gaps in the Sand River case

Gap A: Minimum standards exist, but compliance is not demonstrated at village scale

- The reported shortfall (44% below RDP standards) suggests that the system could not show that households were receiving the minimum.
- Without regular, settlement-level reporting of quantity, distance, continuity, and quality, entitlements become hard to enforce and communities experience them as rhetorical rather than real.

Gap B: A development “need” persists even after policy is established

- Pollard & Walker (2000) identified reliable water as a major development need, and it was confirmed again years later (Maluleke et al., 2005b).
- This persistence indicates that the entitlement was not backed by effective implementation instruments: budgets, O&M systems, oversight, and enforceable performance targets.

Gap C: The entitlement is framed as “minimum needs,” but real household needs are broader

- Even where 25 ℓ/c/d is the entitlement floor, households often require additional water for basic livelihoods (small productive uses). When supply systems do not acknowledge this, the entitlement becomes a floor that still leaves households structurally vulnerable, especially in rural settings.

Gap D: Institutional transitions complicate accountability

- The Sand River experience (from earlier infrastructure notes) includes shifts in who operates and maintains systems (DWAF/DWS → municipality).
- When responsibilities shift without matched capacity, the question “who is accountable?” becomes blurred, and entitlements weaken in practice.

RIDe “Entitlements” for the Sand River catchment

- The Sand River case shows clear formal entitlements (BHNR/Reserve + FBW/RDP minimum service standard), but weak realisation at village level.
- The critical implementation challenge is the institutional split:
- CMA/DWS logic focuses on allocation/protection of the Reserve,
- while municipal WSAs/WSPs carry the practical duty to deliver water to households.
- Where households remain below standards, accountability gaps arise because:
- minimum service compliance is not consistently proven at village scale,
- the entitlement competes politically with large users despite legal priority,
- and institutional transitions and capacity constraints weaken enforcement

Evidence for planning

Key areas can be regarded as evidence for the planning process and are fundamental to operationalising the BHNR.

1) Resources: the BHNR can be “present” hydrologically but absent in practice

a) Paper allocations don’t guarantee accessible supply

Catchment balances and Reserve determinations can “set aside” BHNR volumes, but that does not mean households receive water. In the real world, BHNR is frequently subsumed into broader municipal allocations and system operations. The result: the BHNR exists as a line item, not as a delivered service.

b) Small numbers get treated as negligible

You’ve highlighted a surprisingly damaging issue: BHNR volumes are often tiny relative to irrigation, mining, or large urban/industrial users. In tables (especially in Mm^3/a), BHNR can round to 0.00, which becomes a narrative: “there is no allocation”. That is not just a formatting error—it undermines legitimacy, weakens accountability, and fuels conflict.

c) The ecological reserve “tension point” becomes unavoidable under stress

Once you test scenarios (population growth; higher service standards; drought; reduced yield), the system can tip into deficit. At that point, BHNR, ecological Reserve, and licensed uses become politically entangled. If the BHNR is reframed from 25 l/p/d to 100–110 l/p/d without careful governance design, the water balance can shift from “managed scarcity” to “overt reallocation conflict”.

2) Infrastructure: the missing link between allocation and delivery

RIDe puts infrastructure at the centre because it is the bridge between hydrological availability and real access.

a) NRW and leakage turn BHNR into a “leaky entitlement”

Even if the BHNR volume is increased, a large portion can be lost through non-revenue water, intermittent pressure cycles, illegal connections, failing storage, and deferred maintenance. That means more water is abstracted, yet the constitutional promise is still not met—and the ecological reserve may be stressed in the process.

b) Weak separation of flows enables capture by other uses

Where municipal systems supply households, commerce, and sometimes industry, BHNR water is rarely ring-fenced. In practice, “priority water” can be reallocated within the pipe network (or within municipal decision-making) away from the poorest users—especially when revenue pressures and political demands favour high-volume or paying customers.

c) Operational constraints matter as much as volumes

A municipality can hold an allocation and still fail to deliver because of: pump downtime, energy constraints, poor borehole yields, treatment failures, storage not maintained, or inability to manage intermittent supply fairly. BHNR is not only a volumetric question; it’s an operational reliability question.

3) Demand: population uncertainty and service-level escalation complicate everything

a) “Latest figures” are often contested, and projections amplify error

When base populations are disputed (or outdated), projecting 10–25 years forward compounds uncertainty. The problem is not only statistical—it becomes political: different numbers produce different obligations, different deficits, and different reallocation pressures.

b) Service standards vs BHNR minimum become conflated

A common trap is allowing service aspirations (e.g., 100–110 l/p/d) to quietly replace the BHNR minimum (25 l/p/d) in catchment accounting. When that happens:

- big deficits appear “overnight” on paper,
- allocation conflict escalates,
- and the debate shifts from “minimum core obligation” to “who pays for higher standards”.

c) Demand is not just domestic

Domestic water sits inside wider municipal demand that includes institutions, businesses, and sometimes bulk/industry. If BHNR is not explicitly protected within the demand structure, it becomes vulnerable to higher-volume competitors.

4) Entitlements: who is responsible for what, and how do we prove it happened?

a) Allocation responsibility and provision responsibility are split

CMAs (and national water resource functions) allocate from available water; municipalities (WSAs/WSPs) are responsible for services. That institutional split creates a familiar implementation gap:

- “Allocated but not delivered”

- "Delivered but not protected"
- "Protected in policy but not measurable in performance"

b) Without clear ring-fencing, BHNR becomes administratively invisible

If BHNR is not tracked as a protected volume through municipal planning, budgeting, operations, and reporting, then it is hard to demonstrate compliance—or non-compliance. This is why "zeros in tables" matter: they signal that BHNR is not treated as a first-class entitlement.

c) Conflict is inevitable when scarcity bites—and it will be contested

If deficits deepen, there will be pressure to take water from licensed sectors (agriculture, mining, industry). Those sectors will resist, especially if the state cannot show that:

- municipal systems are efficient (NRW controlled),
- the BHNR is actually reaching intended users,
- and enforcement is consistent and legitimate.
-

Cross-cutting challenges that RIDe highlights

1. Governance credibility problem: reallocate water for BHNR, but lose it to NRW → legitimacy collapses.
2. Accounting problem: BHNR treated as a rounding error → rights become numerically invisible.
3. Risk of perverse outcomes: raising BHNR volumes without safeguards can stress the ecological reserve and still fail households.
4. Mismatch of scales: catchment-level numbers don't translate to village-level access without infrastructure and operational rules.
5. Political economy of scarcity: once deficits appear, the BHNR becomes a trigger for conflict unless the "rules of priority" are explicit and enforceable.

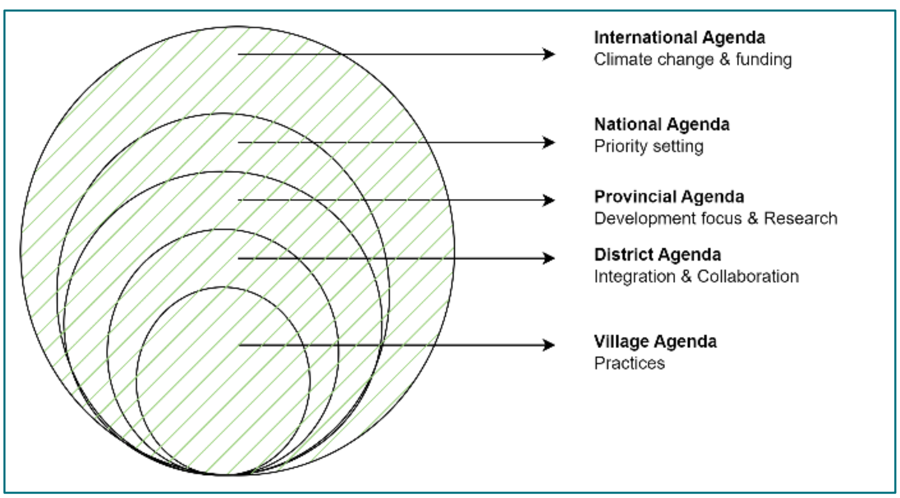


Figure 10 The framework adopted to guide case development for this report and the village case studies (separate document)

Implications of the WRC (2025) rural water supply assessment for operationalising the BHNR

The BHNR is frequently treated as a small volumetric "set-aside" within catchment water balances. The **WRC (2025) assessment** demonstrates why this is not enough: in rural areas, the binding constraint is often **delivery reliability** (continuity of supply), driven by **operations, maintenance, institutional capability, and losses**, rather than the theoretical availability of water alone.

For BHNR purposes, the assessment reinforces a practical truth: **A Reserve on paper does not equal water delivered at the point of use.** Therefore, BHNR credibility depends on showing the full delivery chain: **allocation → abstraction → bulk conveyance → treatment → reticulation → continuity/quality at household level**, with losses and operational risks made explicit.

1) Key BHNR messages from the WRC (2025) assessment

1.1 Access improved over time, but reliability remains weak

The assessment indicates substantial gains in *access* to piped water over the post-1994 period, but highlights that **reliability remains poor and uneven**. Importantly, it notes that reliability data for **Census 2022 were not published** due to **data quality concerns**, limiting the ability to track continuity nationally using that dataset (WRC, 2025: pp. 22–23, 56).

If reliability is not measured and reported, BHNR compliance cannot be demonstrated in practice. Reporting should therefore go beyond annual volumes and include **continuity metrics** (e.g., supply days per year, frequency/duration of interruptions, emergency supply during outages).

1.2 The "service level" target has moved, and BHNR must be kept distinct

The WRC assessment notes that the national service goal shifted from **25 litres/person/day within 200 m** to at least **50 litres/person/day available in the house or yard** (WRC, 2025: p. 22).

This creates a predictable policy trap: service-level targets (progressive realisation) can be mistakenly treated as if they are the Reserve minimum (immediate priority). For defensible BHNR planning:

- **BHNR minimum (Reserve logic)** must remain the protected "first call" minimum; and
- **service standards** must be treated as the target trajectory for municipal planning and regulation.

They should be presented as **parallel lines** in tables and scenarios—not quietly substituted for one another.

1.3 Under-maintenance is structural—and it breaks BHNR delivery

The assessment highlights deep municipal financial constraints in rural WSAs (often district municipalities). It reports:

- a **low ratio of service income to total municipal income** (with much reliance on transfers), and
- chronic under-spending on maintenance relative to the level required to sustain infrastructure.

It references a **National Treasury guideline** (Circular No. 71, January 2014) of **8%** of the book value of Property, Plant and Equipment for maintenance budgeting, and notes reported spending far below this benchmark—median around **0.6%** (WRC, 2025: p. 129).

If maintenance is structurally inadequate, the system will continue to fail intermittently. In that context, increasing BHNR allocations risks producing a perverse outcome: **more abstraction, but not more delivered minimum water** (and potentially greater stress on the ecological reserve).

1.4 Technology choices are not neutral: "big schemes" can create brittle systems

The assessment observes that since local government reforms and Free Basic Water policy, simpler "appropriate" rural options (e.g., **handpumps and spring protections**) have often been treated as interim/second-class, while piped schemes (including large regional schemes) proliferated—often justified by perceived reliability (WRC, 2025: p. 22).

In its recommendations, the assessment argues for shifting away from expensive complex solutions towards **simpler, more affordable options**, favouring **modular, adaptive and decentralised** technologies over large-scale regional schemes for remote rural settlements (WRC, 2025: pp. 8–9).

BHNR actioning should be **technology-portfolio-based**, not single-model. The minimum should be secured through the most **reliable, maintainable, and affordable lifecycle** option for a given settlement pattern—sometimes decentralised options outperform complex networks in real-world sustainability.

1.5 Data gaps undermine long-range BHNR planning (population and reliability baselines)

The assessment notes that Census 2022 did not publish several granular service indicators, including reliability, and it compares population estimates across Stats SA (2011, 2016, 2022) and the DWS Water Services Knowledge System (WSKS), highlighting differences that can materially affect planning and budgeting (WRC, 2025: pp. 56–57).

BHNR calculations (see later) must explicitly state:

- baseline population **year and source**,
- service-level assumptions (25 vs 50 vs higher), and
- scenario ranges (low/medium/high growth), rather than single "escalators".

Where national reliability data are weak, BHNR reporting must rely more on **operational monitoring** (reservoir levels, pump uptime, tanker frequency, DMA/system input volumes, outage logs) and **community monitoring**.

Pathways for meeting the BHNR

The opportunity exists to integrate catchment allocation ('Reserve first'), municipal service provision, and community-level reliability in one plan, leveraging practices under a number of Action Pathways. The project will elucidate pathways that connect national policy with village-level realities. This ensures that rights enshrined in law are translated into practice, and that accountability runs through all levels of governance. Activities are listed under each of the proposed pathways.

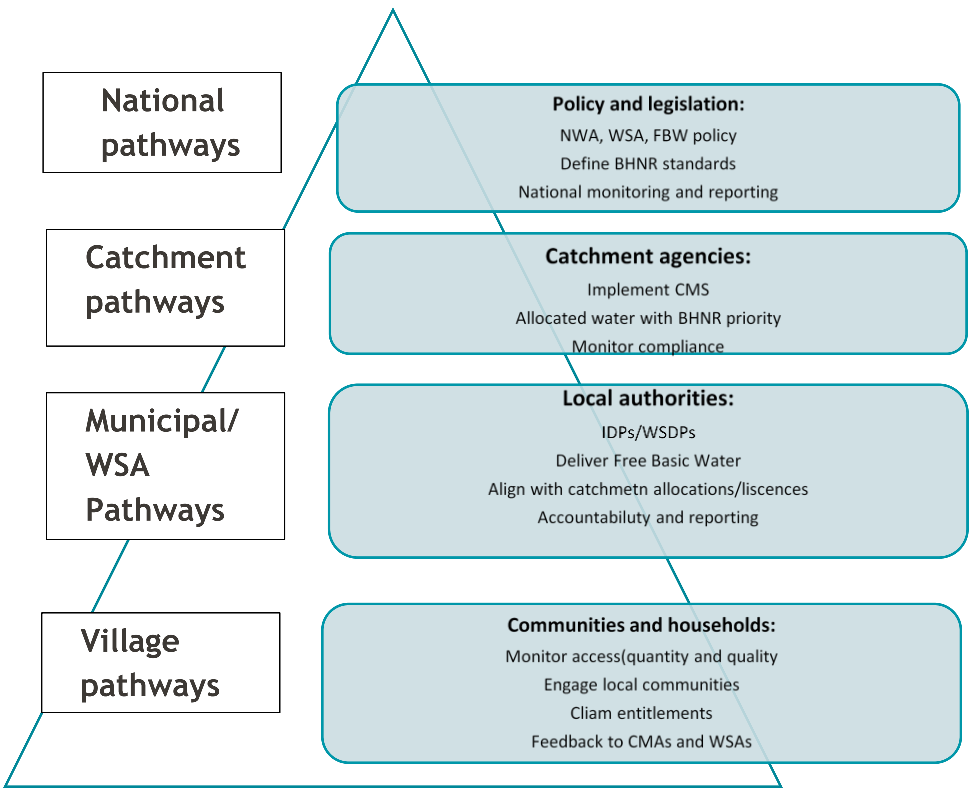


Figure 11 Pathways for Meeting the Basic Human Needs Reserve (BHNR). The diagram illustrates the multi-level process linking national policy and legislation (National Water Act, Water Services Act, Free Basic Water policy) with catchment management agencies

1. National pathways: policy and legislation

One of the key legal instruments to realize the right to water in South Africa is the Reserve, both Ecological and Basic Human Needs. The Reserve is defined in law in terms of the quantity and quality of water, which are required to protect basic human needs and to protect aquatic ecosystems so as to secure ecologically sustainable development and utilization.1 Despite the importance of the Reserve in implementing the right to water, there has been little discussion with regard to how this valuable instrument fits into a rights-based framework.

Using a rights discourse, it will raise some key issues, including, how one enforces and monitors the implementation of the Reserve, how the ecological aspect relates to the human aspect, and the time frame associated with the implementation of such a Reserve and when/if is the government accountable? It must be noted, however, that the purpose of this section is not to provide clear answers to the issues raised, but to initiate a discussion and create awareness around the importance of approaching the Reserve a rights perspective.

South Africa has done much to comply with its Constitutional obligations regarding the right to sufficient water. The Reserve is a key legal tool designed to help achieve the right to water. The definition of the Reserve is clear within the national legal framework dealing with water.

According to the Principle 3, elaborated in the White Paper on a National Water Policy, the foundation for the National Water Act and the Water Services Act, "[t]here shall be no ownership of water but only a right (for environmental and basic human needs) or an authorisation for its use." Furthermore,

Principle 8 defines the Basic Human Needs Reserve (BHNR): "The water required to ensure that all people have access to sufficient water shall be reserved."

Principle 9 elaborates the Ecological Reserve (ER): The quantity, quality and reliability of water required to maintain the ecological functions on which humans depend shall be reserved so that the human use of water does not individually or cumulatively compromise the long term sustainability of aquatic and associated ecosystems. (emphasis added)

Finally, Principle 10 clarifies the legal nature of the Reserve: The water required to meet the basic human needs referred to in Principle 8 [BHNR] and the needs of the environment shall be identified as "the Reserve" and shall enjoy priority of use by right. The use of water for all other purposes shall be subject to authorisation.

1.1 National framing of BHNR across water resource and water services law

National framing of BHNR depends on linking two policy–legal "worlds" that often operate in parallel:

Water resource management (catchment scale): the Reserve concept (BHNR + ecological requirement) and the prioritisation of the Reserve ahead of other uses.

Water services provision (household scale): duties on water services authorities/institutions to ensure minimum service levels, safety, continuity, and emergency provision.

The recurring implementation gap identified in project documents is that the Reserve is treated as a hydrological set-aside, while minimum service delivery obligations are treated as a municipal performance issue—without a systematic bridge between them.

1.1.1 National Water Act: the Reserve, licensing and shortage response (BHNR relevance)

The rights-based approaches synthesis (K5-1512) and the RIDE application draft describe several NWA mechanisms that matter for BHNR actioning:

The responsible authority can grant water use authorisations (licences, general authorisations, or authorisations without a licence), including to organs of state, with licence conditions that may include water resource protection and water management plan requirements.

Licence conditions can be reviewed and amended to prevent deterioration of water resource quality.

In times of water shortage, catchment authorities may limit or prohibit water use, providing an enforcement lever for priorities (Reserve first).

In stressed catchments, compulsory licensing is a potential mechanism to reassess and reallocate entitlements in line with the Reserve.

1.1.2 Water Services Act: minimum standards, oversight and intervention (BHNR relevance)

The WSA framing emphasises that municipalities have duties to ensure water services within the constraints of water resource availability, and must conserve water resources. It also provides a basis for compulsory national standards and for intervention where a water services authority fails to perform required functions.

1.1.3 Free Basic Water policy: a practical planning equivalence for BHNR

Project material repeatedly uses the practical equivalence: 25 litres/person/day ≈ 6,000 litres/household/month (for a typical 8-person household). In practice, Free Basic Water is commonly implemented via municipal indigent policy and tariff design. A national challenge is that the Free Basic Water “basic allocation” is not consistently translated into catchment-level BHNR set-asides and vice versa.

1.2 National guidance on thresholds, monitoring and reporting for BHNR/Reserve

Rights-based documentation stresses the importance of clear monitoring systems and public information. A recurring theme is that monitoring and enforcement responsibilities are ambiguous in practice, and reporting is often not accessible or interpretable at community scale.

National guidance is strongest where it provides:

- Clear minimum service benchmarks (quantity, continuity, flow rate, access distance, water quality).
- Defined emergency supply obligations during outages and drought-related restrictions.
- Mandatory monitoring programmes and reporting to a national regulatory system.
- Performance expectations for NRW and water losses (because these determine whether allocated water reaches households).

1.3 The revised compulsory National Norms and Standards for Water Services (Gazetted 6 June 2025)

The standards also contain additional operational and accountability requirements that are critical for BHNR actioning in practice. These include explicit timeframes for upgrading service levels, access-distance requirements for interim services, record-keeping obligations for tankering, and public communication requirements during drinking-water quality incidents.

1.3.1 Extracted BHNR-relevant requirements from the DWS 2025 Norms and Standards for Water Services

Table 4: BHNR-relevant requirements extracted from the 2025 Compulsory National Water and Sanitation Services Standards

Theme	Practical requirement (summary)
Basic water supply services (formal settlements)	*Within two years of promulgation, the minimum standard is a yard connection/user connection point for existing settlements; basic quantity is 6 kL/household/month at ≥10 L/min; availability ≥358 days/year and interruptions ≤48 consecutive hours; SANS 241 compliance; basic allocation at no cost to indigent households; metering and tariffing requirements apply.*
Interim water supply services (informal settlements)	*Interim service must be provided within 90 days of the WSA becoming aware of the settlement; communal standpipe within 200 m of the furthest household; same minimum quantity, flow rate, continuity and SANS 241 compliance as basic service; zonal metering and record-keeping requirements apply; reliance on tankers should not exceed 12 consecutive months.*
Interruption of water supply services	*Planned interruptions require reasonable notification; if interruption exceeds 48 consecutive hours, emergency supply must be ≥15 L/person/day; alternative supply points must be safe and convenient; extended alternative supply triggers additional monitoring programme registration and reporting requirements (including parameters and record-keeping for tankers).*
Drinking water quality governance	*Water safety plans, monitoring programmes and incident management protocols are required; non-compliance incidents trigger advisory notices and rapid notification obligations; annual drinking-water performance information should be made publicly available and accessible; records to be retained for audit.*
WCWDM, NRW and leak/burst response	*Pressure management requirements; leak repair within 48 hours; pipe burst isolation within 4 hours; NRW and loss performance expectations; 10-year WCWDM strategy and annual plan required, uploaded to the regulatory system.*
Water balance analysis and reporting	*Monthly accounting of abstraction, treatment, supply-zone delivery, consumption and losses/NRW; quarterly reporting of results to the national regulatory information management system.*

1.3.2 Legal status, compliance machinery and enforcement of the 2025 Standards

The revised Standards were prescribed by the Minister of Water and Sanitation in terms of section 9(1) of the Water Services Act, 1997 (Act No. 108 of 1997), as Government Notice No. 6292 in Government Gazette No. 52814 of 6 June 2025, under the short title “Compulsory National Water and Sanitation Services Standards, 2024”. They repeal the 2001 compulsory national standards (General Notice Regulation 509 of 8 June 2001) and bind all Water Services Institutions. Beyond the minimum service parameters summarised in Section 1.3.1, the Standards establish a compliance architecture that converts those minimums into reportable and enforceable duties:

- **Two-year upgrade planning (Regulation 2(9)):** WSAs must submit plans via the national WSDP platform, within two years of promulgation, showing how all consumers in formal settlements will be upgraded to basic services (yard connection / user connection point) — a firm, dated planning trigger for closing basic-service gaps.
- **Annual water services audit (Regulation 26):** each Water Services Institution must include a water services audit in the annual report on its Water Services Development Plan, submitted within four months of the municipal financial year-end. The audit must quantify services by user sector, express service levels as percentages of households, report the water-balance results with total losses and non-revenue water, and evidence O&M expenditure against the original budget — precisely the operational indicators to which Section 1.6 argues BHNR reporting should be tied.
- **Progressive compliance plans (Regulation 27):** where immediate compliance cannot be achieved, a WSA must submit a plan on the Integrated Regulatory Information System (IRIS) within six months of publication detailing how compliance will be achieved progressively — making 'progressive realisation' explicit, time-bound and auditable rather than rhetorical.
- **Offences (Regulation 28):** contravening the prohibitions, failing to issue drinking-water advisory notices, failing to adhere to or provide the plans required under Regulations 2(9), 6(15), 16(9), 20, 26(1) and 27, and disposing of faecal sludge at unauthorised facilities are offences under the Water Services Act's penalty provisions — giving the minimum standards enforcement teeth.
- **Capacity prohibitions (Regulation 11):** municipalities and WSAs may not approve new or additional bulk user connections, or new developments, on water or wastewater systems that lack the capacity to carry the additional load — protecting existing consumers' minimum service from over-commitment, and directly relevant to the demand-escalation risks identified in the evidence-for-planning analysis.
- **Operational viability standards (Part E):** minimum competency, qualification and professional-registration requirements for the heads of the WSA function, planning, provision, and operations and maintenance (Regulation 20); competent maintenance teams, maintenance plans and logbooks for treatment works and networks, benchmarked to national maintenance standards (Regulations 22–23); determination of the actual O&M cost of each water supply system in R/m^3 and a cost-reflective O&M budget per system (Regulation 24); full life-cycle asset management plans and registers included in WSDPs (Regulation 25); and electricity contingency planning for critical infrastructure (Regulation 21). These provisions codify O&M and asset management as regulatory duties — the 'rights-enabling conditions' that the WRC evidence in Section 1.6 shows are chronically underfunded.
- **Incident management protocol (Annexure A):** drinking-water quality failures are classified into Alert Level I (incident), Alert Level II (failure) and Alert Level III (emergency), each with specified response actions, response times, escalation duties and communication requirements — including Boil Water and Do Not Use advisory notices issued through media accessible to the affected communities.

For the BHNR, the significance of this machinery is that the minimum service parameters of Section 1.3.1 are no longer aspirational: they are attached to named plans, national platforms (the WSDP platform and IRIS), fixed timeframes, audit trails and offences. Village- and catchment-level monitoring (Sections 2.7 and 4.5) can therefore be designed to mirror the audit categories that WSAs must already report, allowing communities and CMAs to test official returns against lived delivery.

1.4 Rights principles: where enforceability still tends to break down nationally

Even with updated national standards, implementation gaps persist where rights principles are not fully embedded into enforceable and user-visible systems.

- Participation: processes exist (consultation, forums), but trade-offs and decisions remain opaque, especially when deficits force prioritisation choices.
- Accountability: allocation (CMA/DWS) and provision (municipality/WSA) responsibilities are split; when BHNR is not realised, people struggle to locate a responsible authority for remedy.
- Non-discrimination: informal settlements and remote villages risk systematically lower continuity and slower response to breakdowns; standards exist but enforcement is uneven.
- Transparency: performance data (water balances, NRW, drinking water quality results) is not consistently accessible in formats communities can use.
- Remedy: grievance mechanisms are often unclear, slow, or not linked to enforceable consequences.

1.5 National baseline indicators for evaluating catchment and local practice

A practical baseline should include indicators that can be measured from municipal reporting and verified through community monitoring:

- Minimum service delivery: 6 kL/household/month (or equivalent L/person/day), ≥10 L/min flow at the point of supply, ≥358 days/year availability, and no interruption >48 consecutive hours.
- Emergency minimum: ≥15 L/person/day during interruptions longer than 48 consecutive hours, with documented delivery records (especially for tankering).
- Water quality governance: SANS 241 compliance; water safety plan; monitoring programme; incident protocol; advisories issued and communicated accessibly when required.
- System efficiency: NRW, losses and ILI measured and progressively reduced; leak/burst response times met; monthly water balance completed and quarterly reported.
- Rights and accountability: evidence of accessible complaint channels, response times, and reporting that communities can interrogate.

1.6 National evidence base: the WRC assessment of rural water services (2002–2022)

The WRC national review of rural water supply and sanitation services (Still, Eales, Jula & Gibson, 2025; WRC Report No. 3206/1/25) provides the evidence base against which national BHNR pathways must be planned. The review focuses on the 21 Category C2 district municipalities — rural district WSAs whose sole function is water and sanitation services and which jointly serve 65% of South Africa's rural population. Its findings confirm the central premise of this report: the right to water depends on functioning systems, not infrastructure counts.

Key findings

- **Access improved, reliability did not keep pace.** Rural piped-water access rose from 44% (1994) to 74% (2022), yet only 48.4% of rural households with piped supply reported a stable supply in 2011, improving marginally to 51.7% in 2016. Stats SA withheld the Census 2022 reliability results due to data-quality concerns.
- **C2 WSAs are financially distressed and grant-dependent.** The median ratio of service income to total income is 10%, with roughly 90% of income derived from national transfers; the typical C2 municipality collects only about 60% of what it bills.
- **Maintenance is chronically under-funded.** Median reported maintenance expenditure was 0.6% of the R82.2 billion asset book value under C2 management — far below the Treasury guideline of 8%.
- **Liquidity is deeply negative.** The 21 districts jointly carried R11.2 billion in current liabilities against R4.5 billion in cash (2021/22); excluding four healthier districts, the remaining 17 held R10.1 billion in liabilities against R2.5 billion in cash, deferring maintenance and supplier payments as a result.
- **Treatment performance is weak at scale.** The C2 districts recorded a median Blue Drop score of 54.8% (2023) and a median Green Drop score of around 38% (2021/22).

Implications for national BHNR framing

Translated into the national pathway of this report, the assessment implies that the BHNR must be framed not only as a volumetric entitlement but as a deliverable minimum service, evidenced through continuity and access measures. Specifically:

- **Codify the separation of concepts:** the BHNR (Reserve minimum) versus service standards (progressive targets) — reinforcing the distinction developed in Sections 1.1–1.3.
- **Tie BHNR reporting to operational indicators** — continuity of supply, emergency provision and water-quality compliance — not just allocation volumes, in line with the 2025 Norms and Standards requirements extracted in Section 1.3.1.
- **Elevate O&M and asset management as core rights-enabling conditions,** preventing a "new-build bias" in which capital programmes expand while existing systems fail. The WRC review recommends that at least half of all future capital grant funding be directed to capital maintenance, so that infrastructure remains in a constant state of planned renewal.
- **Address the grant architecture:** per-capita operations grants vary greatly across C2 districts, and the review argues for targeted increases to the least well-endowed districts rather than across-the-board rises.

Key BHNR-relevant benchmarks:

- Basic minimum service quantity: 6 kL/household/month (basic + interim).
- Minimum flow rate: at least 10 L/min.
- Continuity: at least 358 days/year; no interruption longer than 48 hours.
- Emergency minimum during interruptions: at least 15 L/person/day (when interruption exceeds 48 consecutive hours).
- NRW/water loss performance expectations: NRW 20–30%, losses 10–20%, Infrastructure Leakage Index (ILI) 2–4; per capita usage benchmark 120–180 L/c/d.
- Monthly water balance and quarterly reporting requirements to the national regulatory information system.

2. Catchment pathways: regional water resource management

Since the catchment is the unit on which management of water resources rests, planning for meeting the BHNR cannot be adequately done without attention to water resources planning and management process at the level of the catchment. The processes of democratic decentralization associated with WRM in South Africa takes the CMA as kingpin in planning and allocation for a human rights approach over the long term. And although the CMA is not directly responsible for operationalising the reserve, (a responsibility of DWAF national) it needs to recognize that a rights approach to water is central to water supply and management.

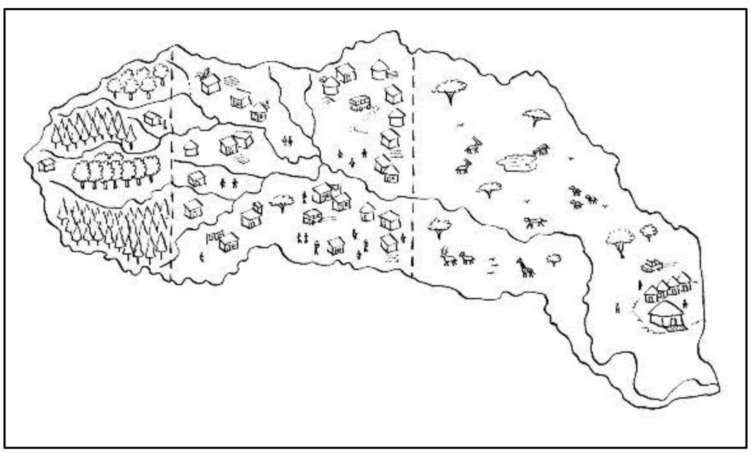


Figure 12 Water moves through a catchment towards the sea with shifts in allocation obligations as it flows

Planning for a rights approach at the catchment level

The CMA cannot allocate and issues licenses for water use until it has shown that the reserve is accounted for. In this regard the CMA has an essential role to play in recognizing the reserve as a right and with the highest priority in the water allocation process. Furthermore, the CMA has a crucial role, according to the NWA, to ensure that longterm sustainability underpins the management of water on a catchment basis.

Although the reserve falls under the direct control of the Minister (NWRS p 16) it is the CMA that is tasked with the responsibility of ensuring that water resources are managed in order to provide for the reserve. This includes recognising the status of the reserve in the Catchment Management Strategy, monitoring compliance, and drawing up allocation plans that give the highest priority to the reserve. Since one of the initial functions of the CMA (NWA, Section 80 a) is "to investigate and advise on the protection, use, development, conservation, management and control of water resources management in its WMA", it is a key role player in contributing to sustainable water use practices and water security at the catchment level. The key instrument in this regard is the ecological reserve and is accorded the status of "right" which means that all the concepts that apply to the BHNR apply equally to the ER.

Notes from the field

The water resources consultant to the IUCMA, who says that when they plan for the basic human needs reserve within the water allocation process at the IUCMA, they bundle the BHNR requirements calculated at 25 liters per person per day into the ecological reserve and hope that that is going to be an adequate provision for the municipalities. This calculation was projected over immediate 5- and 10-year provisions by Patsy Schurmann, from IWR at Rhodes University for the calculations earlier on. They do not factor in that much of that allocation to the municipalities could end up going to commercial, industrial, and diversional fire uses other than the basic human needs reserve and multiple-use provisioning. They also don't really deal with boreholes and off-belt supply systems. But this needs a closer look at, and part of the process will be to establish a meeting with the IUCMA to see exactly how these calculations can be formalized, and follow-up meetings with the IUCMA will be part of the process.

1) Current practice at the IUCMA for the BHNR

- During allocation planning, the **BHNR requirement** (calculated at **25 litres per person per day**) is reportedly **bundled into the Ecological Reserve**, with an assumption that this will be an adequate provision for municipalities.
- Earlier BHNR calculations and **5- and 10-year projections** were reportedly prepared by **Patsy Schurmann (IWR, Rhodes University)** and used in earlier work.
- The approach may not explicitly account for:
- municipal water that is used for **commercial, industrial, diversional, and fire-fighting** purposes (beyond BHNR intent),
- **boreholes** and **off-belt supply systems** and the realities of reliability/quality/operations.

Status: These points are working observations and must be confirmed against IUCMA's current documentation, model assumptions, and operating rules.

2) "Bundling" the BHNR with the Ecological Reserve (risk statement)

If BHNR is "counted" within the Ecological Reserve without clear ring-fencing and delivery logic, there is a risk that:

- BHNR is treated as "covered on paper" while **actual municipal/basic-needs availability is overestimated**,
- municipal allocations are effectively diluted by **non-domestic uses**, undermining equity and multiple-use provisioning,
- groundwater/off-belt supplies are **misrepresented** (assumed reliable when they may be intermittent or constrained),
- planning decisions and licensing outcomes become difficult to defend because the BHNR provision is **not clearly formalised, traceable, or auditable**.

3) Key questions for IUCMA

A. Accounting and method

- Where does BHNR appear in the allocation framework/model: **explicit line item** vs **embedded within ecological reserve** vs narrative only?
- What is the current BHNR standard used (still **25 l/p/d**?), and what population datasets and growth assumptions underpin **5- and 10-year** projections?
- What is the rationale/policy basis used internally for the adopted l/p/d standard and projection method?

B. If "bundling" is used

- How is the BHNR component **protected** when trade-offs are made (e.g., during drought restrictions or ecological constraint periods)?
- How does the bundled amount translate into **municipal supply decisions** or enforceable operating rules?

C. Municipal demand reality

- Does IUCMA segment municipal demand (basic domestic vs productive vs commercial/industrial vs fire-fighting), or treat it as one block?
- What assumptions are used about how much of "municipal" water actually reaches **basic human needs** and/or **multiple-use** outcomes?

D. Groundwater and off-belt systems

- Are boreholes counted as part of BHNR provision? If yes, is this based on **assured yield and functionality**, or nominal infrastructure presence?
- How are off-belt supplies represented (transfers, local schemes, non-river abstractions), and how is **reliability/quality/seasonality** handled?

E. Formalisation and transparency

- Which documents govern this practice (reserve determination reports, allocation plan/reconciliation strategy, model inputs/spreadsheets, operating rules)?
- What is the evidence trail showing how BHNR is secured in planning and allocation decisions?

4) What we need from the first meeting (deliverables)

1. A clear explanation (and ideally a simple diagram) of **how BHNR is represented** in the allocation process.
2. A list of **assumptions** used (population, l/p/d, growth, demand segmentation, supply reliability).
3. Identification of **known limitations** and how IUCMA currently mitigates them.
4. A document/data access plan: what IUCMA can share (and under what conditions).
5. Agreement on follow-up engagements with the IUCMA.

2.1 What the catchment pathway is expected to deliver for BHNR

At catchment scale, the “BHNR” does not mean that CMAs directly provide household water. Rather, it means ensuring that the Reserve (including BHNR) is set aside first in allocation decisions, that operating rules and restrictions protect that priority during shortages, and that monitoring and enforcement prevent the Reserve from being undermined by unlawful or excessive use.

2.2 Key BHNR-relevant levers in catchment governance

1. **Catchment strategies and allocation plans**: CMSs should include strategies and measures for the protection, use, development, conservation, management and control of water resources, including monitoring of water resources and water use.
2. **Licensing and licence conditions**: licence conditions can require water management plans and protect water resources; conditions can be reviewed/amended to prevent deterioration of water quality and to uphold priorities.
3. **Shortage management**: in times of shortage, catchment authorities can limit or prohibit water use; drought restrictions should explicitly prioritise domestic minimum supply and the ecological requirement.
4. **Compulsory licensing in stressed systems**: where a catchment is ‘closed’ or over-allocated, compulsory licensing is a possible tool to reassess and reallocate entitlements consistent with the Reserve.
5. **Information systems**: national/catchment monitoring systems should make water resources information accessible and usable by the public and stakeholders.

2.3 Practical challenges (RIDe perspective)

1. Data and projection uncertainty: contested base population figures and 10–25-year projections amplify error and political contestation.
2. Rounding-to-zero optics: BHNR volumes are small relative to large users; when reported in large units they can appear as 0.00 in water balance tables, undermining legitimacy and accountability.

		From Table 6.1 Summary of the Reserve (groundwater component?)							
Biophys. Node (Quarter Catch)	Flow site EWR site	Groundw Recharge MCM	Populat. (estimated)	Baseflow million m^3/a	Low flows EWR-MLF	BHN Reserve	Reserve MCM	Groundw Use MCM	Stress Index %
X23A	X2H010	10.69	28783	1.71	4.79	0	4.79	0.07	1
X23B	X2H080 ?	12.38	28738	3.18	5.13	0	5.13	0.65	5
X23C	X2H024	6.98	28738	3.34	2.93	0	2.93	0.12	2
X23D	X2H031	12.89	28783	2.43	6.11	0	6.11	0.17	1
X23E	X2H008	12.02	28738	3.18	5.32	0	5.32	0.18	1
X23F	X2H083 ?	20.29	53913	1.63	8.5	0.492	8.99	2.41	12
X23G	X2H088 / EWR K7 ?	11.2	598	2.24	4.97	0.005	4.98	0.26	2
X23H	X2H22 / EWR K7	11.2	3837	1.92	10.78	0.035	10.82	0.66	4

3. Escalation pressures: raising domestic allocations (e.g., from 25 L/person/day to higher service targets) can push catchment balances into deficit, triggering reallocation conflict and likely resistance from agriculture and industry.

4. Ecological Reserve tension: if domestic allocations increase without parallel loss reduction and demand management, the trade-off between BHNR delivery and ecological protection becomes unavoidable and often implicit.

5. Allocation vs provision split: CMAs allocate from available water; municipalities provide services. Without explicit 'handover' mechanisms, BHNR remains a paper set-aside.

2.4 Making BHNR visible and defensible in catchment water balances

1. The recorded statement emphasises that "zeros in tables" are not merely a presentation issue: they are read as evidence that BHNR has not been allocated. Catchment reporting conventions should therefore prevent BHNR from disappearing.

2. Use ML/year, m^3/day, or L/s (operationally meaningful units) when reporting BHNR and Reserve requirements.

3. Increase decimal precision for Reserve components if reporting in Mm^3/year.

4. Include a BHNR proportion indicator (BHNR as % of yield/available water) in all summary tables.

5. State explicitly whether BHNR is net (delivered) or gross (loss-adjusted at source).

2.5 Compliance monitoring and enforcement: from rules to consequences

Rights-based documentation stresses that the state has a duty to ensure compliance with the Reserve. This includes refusing, amending or revoking entitlements that contribute to violations of the Reserve, and ensuring dispute-resolution mechanisms. Civil society and private actors can support monitoring, but enforcement authority remains with the state.

Practical mechanisms that can be strengthened include:

1. Licence compliance audits linked to Reserve compliance indicators (including drought restrictions and operating rules).

2. Clear triggers for directives and restrictions when monitoring shows Reserve risk (and public communication of the triggers).

3. Transparent processes for reallocation debates in deficits, including socioeconomic and equity considerations.
4. Publicly accessible reporting dashboards or summary reports that show BHNR and ecological requirement status over time.

2.6 Linking village-level evidence to catchment decisions ('catchment-to-village' pathway)

1. A recurring gap is the absence of a routine mechanism for village realities to influence catchment decisions. To test and institutionalise such linkage, the pathway should combine community monitoring (service outcomes) with catchment data (resource status and allocations).
2. Standardise a small set of community indicators (days with supply; interruptions >48h; emergency supply; distance to source; functionality of communal points) and align them to national standards.
3. Create a periodic reporting channel from municipalities and community structures into CMA forums/CMS review processes (e.g., quarterly or biannual).
4. Use RIDe diagnostics in catchment deliberations to distinguish resource scarcity from infrastructure/operations failure.
5. Where catchment deficits exist, require that NRW reduction and municipal efficiency plans are tabled alongside any request for increased domestic allocation.

2.7 Treating the BHNR as deliverability: implications of the WRC (2025) assessment

The WRC assessment shows that catchment water balances can be "in balance" while households experience chronic supply failure: the allocation exists on paper, but distressed municipal systems cannot convert it into a reliable minimum service. Because BHNR volumes are small relative to large users, the BHNR line item can also become politically invisible when rounded down in allocation tables. For catchment planning this implies:

1. **Require a delivery-risk note alongside every BHNR allocation:** is the downstream municipal infrastructure able to convert the allocated volume into a reliable minimum service? Allocation without deliverability assessment is not BHNR compliance.
2. **Present the BHNR in units that do not disappear** (ML/a, m^3/d or L/s), with precision conventions for priority reserves so the minimum is never rounded to zero in catchment balance tables.
3. **Create a feedback loop from village to catchment:** village-level reliability data must inform catchment prioritisation during scarcity, rather than relying only on aggregated annual volumes — the 'catchment-to-village' pathway of Section 2.6 running in reverse.

3. Service delivery pathways: municipal and Water Services Authorities

Municipal systems convert catchment allocations into household access. In the recorded statement, two risks dominate: (i) increased allocations can be lost to non-revenue water and leakage, and (ii) domestic priority water can be absorbed by other uses supplied through municipal networks (commercial/industrial or higher-volume users).

3.1 District planning response: Bushbuckridge Local Municipality (RIDe synthesis)

Context from the interview notes

Mpho Makhuvhu (formerly scientific manager at Bushbuckridge Water Board, from 2012) is now Deputy Director: Technical Services at Bushbuckridge Local Municipality, responsible for laboratory services and operations at the Bushbuckridge water works. The key institutional arrangement described is a rights-of-use contract / service level agreement (performance-based and reviewed on a five-year cycle) for operating a treatment plant owned by the Department of Water and Sanitation (Infrastructure Branch).

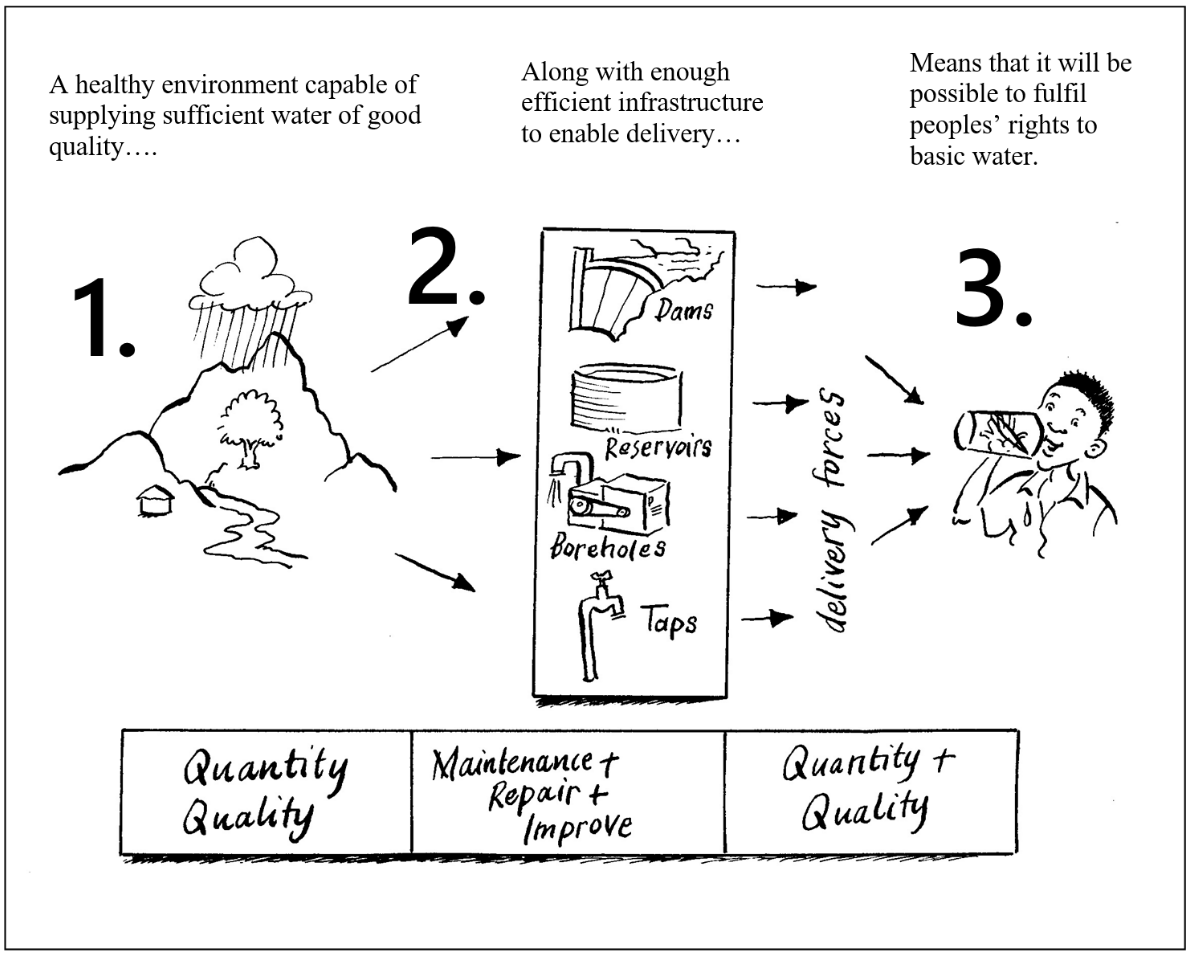


Figure 13 The links between catchment resources, infrastructure and demand at taps

Resources

The Inkomati-Usuthu CMA (IUCMA) is reported to have authorised an abstraction licence of approximately 116 megalitres per day. This authorised volume is understood to include domestic, commercial, and industrial uses. The interview notes that IUCMA has indicated no additional water will be authorised, shifting the practical emphasis to water conservation and demand management within existing authorisations.

Additional district concern recorded in earlier interview notes: the CMA water balance work was described as bundling BHNR into the Ecological Reserve (i.e., assuming ecological protection automatically covers basic human needs), which in practice may not account for municipal system losses, unlawful connections, and local irrigation/productive use.

RIDe implications (Resources):

- Treat BHNR as a protected minimum within the authorised volume and make the BHNR share explicit in planning and reporting.
- Separate “domestic minimum” from additional discretionary municipal supply (including commercial/industrial growth) to avoid silent crowding out of BHNR.

Infrastructure

The interview describes a major upgrade in 2014 from about 50 ML/day to about 100 ML/day at the Inyaka Dam works, and current planning for an additional upgrade of about 15 ML/day towards an intended capacity of about 150 ML/day. Bushbuckridge also operates numerous decentralised package plants (reported as 14), which are described as expensive to operate and maintain relative to their output. A key operational concept raised is the difference between “manufactured demand” (what plants can produce) and “supplied demand” (what can be delivered), constrained by reticulation limitations and losses.

System losses and NRW (as reported):

- Non-revenue water reported as very high (around 90%), largely linked to low payment levels, unbilled authorised consumption, and system management challenges.
- Physical losses reported around 42% (from an IWA water balance conducted by a consultant).
- Loss drivers include vandalism, unlawful connections, overflows, and leakage.
- RIDe implications (Infrastructure):
- Use measured losses (not assumed percentages) when converting BHNR delivered requirements into system input requirements.
- Prioritise reticulation rehabilitation and pressure management where these are the binding constraints on supply, even when treatment capacity exists.
- Evaluate centralisation versus package plants using a life-cycle cost and reliability lens (O&M intensity, spares, operator skills, energy, and chemical costs).

Demand

Planning assumptions reported in the interview include: population estimates using local/administrative data (including GIS data as recent as 2025) with an average of four people per household, yielding a municipal population estimate just over one million people. The planning norm referenced is about 100 litres per capita per day, reflecting a transition from basic supply to higher domestic service levels (toilets, showers) and livelihood-related uses. The interview also notes continuing unrecorded abstraction by users directly from rivers in some cases.

Earlier Bushbuckridge interviews (Mpho Makhuvhu) also report a higher domestic planning provision of about 150 l/person/day, justified by a shifting household service level (showers, toilets, baths) and routine productive use (e.g., small gardens). Those interviews further indicate a loss allowance of about 10% on the 150 l/person/day (approximately 15 l/person/day), and acknowledge gaps in tracking/billing commercialised users and unauthorised connections. Taken together, the interviews suggest that municipal planning assumptions vary (100–150 l/person/day) and should be reconciled explicitly for CMA engagement.

Simple volumetric check:

1. If the population is 1,000,000 and the planning norm is 100 l/person/day, then total demand is 100,000,000 litres/day (100 ML/day).
2. If 42% represents physical losses on the system, then a delivered requirement of 100 ML/day would imply a system input of about 172 ML/day (100 ÷ (1−0.42)).
3. If the 100 l/person/day figure is instead being treated as a production target at the works, then delivered volumes would be lower once losses and operational constraints are applied.

RIDe implications (Demand):

- Maintain separate demand categories: BHNR minimum, higher domestic service level, productive use, and commercial/industrial. Do not hide these in a single per-capita figure.
- Treat unauthorised connections and direct river abstraction as demand components requiring estimation ranges and an enforcement/regularisation strategy.
- Where population figures are disputed, carry explicit low/medium/high scenarios and show the consequences for volumes, costs, and service levels.

Entitlements and implementation mechanisms

Enforcement and implementation mechanisms described include municipal bylaws (noted as weakly enforced), a stakeholder hotline ("*Legothla*") to encourage conservation and awareness of bylaws, and a revenue recovery strategy focused on commercial and large users. Measures under consideration include smart meters, a "quick wins" programme targeting about 60 high-volume commercial customers, and prepaid metering. The interview notes the use of a DWAF "water utility model" principle: revenue recovered should be reinvested into operations and maintenance.

RIDe implications (Entitlements):

- Use service level agreements and rights-of-use contracts to define minimum BHNR delivery performance metrics (reliability, interruptions, water quality, reporting).
- Strengthen the enforceability of bylaws and link enforcement to metering and customer management (especially for productive/commercial use within domestic areas).
- Make clear distinctions in reporting between: authorised unbilled consumption, unauthorised consumption, and physical losses.

3.2 Cross-scale alignment: key mismatches and RIDe-based actions

The interviews show a recurring mismatch between catchment-scale allocation assumptions and municipal service delivery realities. RIDe helps convert this into a practical alignment agenda.

Table 5: Cross-scale alignment of RIDe elements: national/catchment framing versus Bushbuckridge district framing

RIDe element	Typical national / catchment framing (from interviews)	Bushbuckridge district framing (from interviews)
Resources	*BHNR minimum norm used for domestic planning in allocation models; risk of BHNR being implicit or bundled with other Reserve categories.*	*Authorised abstraction (approx. 116 ML/day) must cover domestic plus commercial/industrial; no further water expected.*
Infrastructure	*Policy intent often not translated into loss-adjusted system input requirements and measurable performance reporting.*	*Treatment capacity upgrades (50→100 ML/day; planned 150 ML/day) plus costly package plants; binding constraints include reticulation and high losses/NRW.*
Demand	*Single per-capita minimum can dominate models; population assumptions may be inconsistent across institutions.*	*Planning norm around 100 l/c/d; population estimated >1 million; productive use and unauthorised use influence real demand.*
Entitlements	*Need clearer accountability and standard reporting between CMA and WSA; ring-fencing of BHNR often weak.*	*Rights-of-use/SLA governs operation of DWS-owned plant; bylaws exist but enforcement is weak; move toward smart meters and targeted billing to restore O&M capacity.*

RIDe-based actions suggested by this evidence:

1. Joint CMA–WSA scenario planning that explicitly shows BHNR delivered minimum, loss-adjusted system input, and the remaining envelope for other municipal uses.
2. A shared measurement and reporting protocol (bulk meters, WTW logs, NRW and IWA water balance updates) to reduce disputes about "allocation versus deliverability".
3. Explicit treatment of productive and commercial use within domestic areas (regularisation, tariff/billing, and bylaw enforcement).
4. A prioritised NRW reduction and network rehabilitation plan that links funding, performance targets, and service level improvements.
5. Clear shortage rules that protect BHNR first (and define how restrictions will be applied and communicated).

3.3 Planning for the Basic Human Needs Reserve (BHNR)

RIDe Synthesis of Bushbuckridge Interviews

This section distils two interviews into two planning responses (National and District) for implementing the BHNR. The analysis is structured using the RIDe framework, which helps link catchment-scale allocation and policy intent to municipal service delivery realities at scheme and household level.

Interviews synthesised:

- National interview: national / provincial-level water services planning and institutional arrangements for BHNR implementation (policy-to-practice interface).
- District interview: Bushbuckridge Local Municipality water services (Mpho Makhuvhu- Deputy director Technical services BBR LM) focusing on operational planning, rights-of-use, and system performance.

WSA planning response for BHNR (RIDe synthesis)

Resources

National and catchment planning practice commonly anchors BHNR domestic demand to a minimum norm (historically expressed as 25 litres per person per day). This norm is useful for defining a minimum entitlement, but on its own it does not describe the system input needed to deliver that minimum in real municipal systems with losses, intermittency, and varying levels of service.

Implications for regional and national planning (Resources):

- Keep BHNR explicit as a distinct planning line item (do not let it disappear inside other categories).
- Require scenario-based planning where service levels are higher than the minimum norm (e.g., 50–100 l/c/d), showing the trade-offs and augmentation needs.
- Clarify how BHNR is represented in CMA water balances when the Ecological Reserve is also being set (avoid assumptions that one automatically covers the other in practice).

Infrastructure

The national gap highlighted in the interviews is the step from policy intent to practical intent. A RIDe lens makes infrastructure and system performance central: the BHNR at the point of use depends on treatment, conveyance, reticulation condition, operations, and metering.

Implications for national planning (Infrastructure):

- Promote a “two-number” BHNR calculation: (1) the delivered minimum and (2) the loss-adjusted system input required to reliably deliver it.
- Specify minimum monitoring points (bulk offtake, WTW inlet/outlet, key reservoirs, and selected district metered areas) to separate losses, authorised unbilled use, and illegal use.
- Align BHNR expectations with WSDP and budget processes so that municipalities plan for reliability, downtime, and asset condition, not only for average volumes.

Demand

National guidance is experienced by WSAs as uneven, particularly where population baselines are uncertain and where domestic systems are also supporting productive uses (multiple-use reality). RIDe suggests that a single per-capita norm is insufficient for planning where service levels and use profiles are changing.

Implications for national planning (Demand):

- Require WSAs to declare population sources and growth assumptions (and where they apply local adjustments).
- Encourage demand segmentation: minimum domestic (BHNR), higher domestic service levels, productive use, commercial/industrial, and unaccounted-for use.
- Link demand management to measurable loss reduction and enforcement, not only to messaging.

Entitlements and accountability

At national level, the interviews point to a governance and accountability gap: policy intent is clear, but practical obligations, monitoring responsibilities, and escalation pathways are not always operationalised across the CMA–WSA interface.

Implications for national planning (Entitlements):

- Clarify who is accountable for what along the chain: CMA (protection of Reserve and allocation rules) versus WSA (service delivery and reliability).
- Promote ring-fencing of BHNR in municipal accounting and reporting so that minimum domestic supply is protected during shortages and not absorbed by NRW or discretionary uses.
- Provide standard templates for BHNR reporting (delivered volumes, interruptions, losses, key risks) to support comparable oversight across WSAs.

3.4 Case study 1: Bushbuckridge local municipality

It is important to note here that the BBRLM as status as a WSA (in line with the Mopani District Municipality as a WSA)

Table 6: Key quantitative figures cited in the BBRLM interview

Item	Interview note value (to verify)
WTW upgrade (2014)	Approx. 50 ML/day to 100 ML/day. Planned 50ML/d upgrade
Planned additional upgrade	*immediate. +15 ML/day; target capacity stated as ~150 ML/day*
Number of package plants	*14 decentralised package plants*
Authorised abstraction (IUCMA)	*Approx. 116 ML/day (all uses combined)*
Population planning figure	*Just over 1,000,000 (based on local/ONGIS data; 4 people/household)*
Planning per-capita norm	*Approx. 100 l/person/day (reflecting higher service levels and livelihood use)*

Item	Interview note value (to verify)
Physical losses	*Approx. 42% (IWA water balance by consultant)*
Non-revenue water	*Approx. 90% (linked to non-payment and unbilled authorised use)*
Key loss drivers	*Vandalism, unlawful connections, overflows, leaks*
Revenue recovery interventions	*Smart meters; prepaid meters; target ~60 large users for early billing culture*
Engagement mechanism	*Hotline and communication "Legotla" for conservation and bylaw awareness*

Additional WSA notes (Bushbuckridge): ward-level monitoring, borehole strategy, and Innovation Africa coordination (RIDe lens)

1. **Demand / information:** The municipality maintains a ward-level "water supply status" analysis (per ward) and can update it weekly as conditions change, supporting more responsive demand and service planning.
2. **Infrastructure (boreholes as alternative supply):** Boreholes are currently treated largely as an alternative/stop-gap to the bulk system. However, there is an active intention to rationalise and ultimately decommission many boreholes because they are expensive to maintain—particularly due to fixed electricity charges that can accrue even when boreholes are non-functional.
3. **Entitlements / governance (handover and future use):** A key planning question is whether there is a formal handover plan for boreholes once bulk supply improves: options include (i) full decommissioning and transfer to villages/communities for locally managed productive uses; (ii) co-management agreements for continued alternative supply and shared maintenance; and/or (iii) retaining selected boreholes as emergency back-up sources to cover bulk-system outages (source switching as conditions require).
4. **Institutions / coordination (Innovation Africa):** The Innovation Africa case was raised as a governance and coordination risk: clearer understanding is needed of the build-operate-manage model and roles. Weak coordination has reportedly created tensions, including instances where engagement routed directly via traditional authorities has generated friction with communities and local leadership structures.

Innovation Africa **and non-municipal sources: O&M assimilation, user payments, multiple-use pressures, and NRW/WCDM priorities (RIDe lens)**

1. **Infrastructure / institutions (O&M assimilation and asset governance):** A key issue is that operation and maintenance (O&M) for Innovation Africa installations was funded during the project phase; after programme withdrawal, O&M needs to be absorbed into the Bushbuckridge municipal financial plan. This is difficult where assets are not formally negotiated, accepted, and recorded in the municipal asset register, creating uncertainty about responsibility, budgeting, and maintenance frequency.

2. **Demand / behaviour (user charges and payment culture):** Tensions arise where Innovation Africa installations charge approximately R10 per household per month, while nearby municipal boreholes are free to users. This undermines willingness to pay and reinforces a broader culture of non-payment and weak contribution norms, complicating sustainable O&M for community supplies.
3. **Infrastructure (energy options):** Solar pumping is seen as a potential mitigation for operating costs, but it still requires a functioning O&M model, clear roles, and predictable financing—especially for systems operating outside municipal management arrangements.
4. **Demand / entitlements (potable bulk water used for productive/commercial farming):** A further pressure point is use of potable bulk supply water for small commercial farms without payment. The municipality intends to identify these users and negotiate compliance—either through full payment for potable water, or preferably by shifting commercial uses to alternative sources (e.g., boreholes/raw water supplies) rather than diverting treated domestic water.
5. **Institutions (cross-sector coordination for multiple-use solutions):** There is an expressed need for closer coordination with the Department of Agriculture to identify alternative water options for productive uses. This reinforces the importance of recognising and planning for multiple-use realities rather than treating them as 'illegal add-ons' to domestic supply.
6. **Infrastructure / finance (NRW, losses, and WCDM programme options):** Non-revenue water (reported at ~90%) and physical/system losses (reported ~45%) are identified as priority issues requiring better quantification. Options under discussion include a PMU-supported study and use of a portion of the Municipal Infrastructure Grant (e.g., 10%) to develop and implement a Water Conservation and Demand Management (WCDM) strategy.
7. **Institutions / support (technical assistance for NRW actioning):** An additional option is bringing in MISA support for a non-revenue water 'actioning' programme aimed at reducing losses and increasing revenue through improved metering and payment for potable water use.

3.5 Case study 2: Mopani District: BHNR allocation and delivery planning

This case documents how Mopani District approaches allocation planning in relation to the **Basic Human Needs Reserve (BHNR)**, based on the district interview. The discussion focused on whether BHNR (e.g., **25 litres per person per day**) is translated into explicit allocation calculations, or whether allocations are primarily managed through treatment plant capacity, measured flows and operational controls.

How allocation is currently operationalised: capacity, meters, and controls

The interview with WSA staff suggests that Mopani District does not typically operationalise BHNR allocations through a simple per-capita calculation (e.g., population × 25 l/c/d × losses). Instead, allocation is managed primarily through three interlinked elements:

- **Treatment plant design and capacity information**: the district relies on the design specifications/capacity of water purification plants as the baseline for system supply potential.

- **Volumetric monitoring using flow meters** : the district uses meters to determine volumes taken from different storage tanks and transferred into specific areas.
- **Operational control and pump operations**: allocation across areas is managed through operational regulation—pump operators and system controls that influence how water is distributed to different supply areas.

In practice, this reflects an **operations-led allocation model**, where the “allocation” is understood through what the system can produce and what is measured as being delivered, rather than through a BHNR-first demand calculation at community level.

Demand estimation: per-capita norms not readily stated or applied in routine practice

A key line of questioning explored whether Mopani calculates demand explicitly using population and a per-capita benchmark (25, 55, 100 l/c/d etc.), and then compares this against treatment plant capacity to determine whether the system is under- or over-capacity.

The interview responses indicate that this type of per-capita demand calculation is **not readily available in the operational process**. When asked what per-person number is used, the response was that **“it depends”**, and the respondent indicated that they would need to consult internal documents before sharing the full information that informs district responses to demand. Discussion pointed towards operations-led supply, meaning this depends on the capacity of the WTW outputs.

This indicates a potential gap between BHNR as a normative requirement and the **extent to which BHNR (and per-capita demand figures more broadly) is embedded as a routine planning and allocation tool** at village or supply-zone level.

Multiple sources of supply complicate BHNR accounting

The interviews highlighted that supply is not always exclusively from treatment plants. The district noted that some areas also receive water from **boreholes**, meaning that not everyone drawing water in a system is supplied only through a single treatment plant.

This multi-source reality complicates BHNR allocation planning and accounting because:

- the relationship between a plant’s output and the population served is not always direct; and
- compliance with BHNR at community level cannot be inferred from treatment plant production alone without clearer supply-zone mapping and accounting.

Planning approach and collaboration: improving BHNR allocation needs to be grounded in district systems

In the same discussion, the district strongly emphasised that planning and model development should be done **in close collaboration with the DWS district office**, rather than through external teams (donors, researchers etc) working independently or ‘moving too quickly’ (noting concerns that have emerged in other contexts).

The interview indicated that improved planning would need to align to district systems and instruments, including:

- **asset management** information;

- district planning instruments (including WSDP-related information, as referenced); and
- development of feasible **financial and operational models** that fit district conditions.

Key gaps revealed by the interview

The interview points to several gaps that are relevant for BHNR planning and reporting:

- **BHNR is not clearly translated into explicit per-capita demand calculations** in routine allocation discussions.
- The district appears to rely heavily on **volumetric measurement and operational control**, which may not provide sufficient evidence of whether BHNR is met across specific communities/supply zones.
- **Multi-source supply** (treatment plants plus boreholes) further complicates the ability to assess demand, allocation, and BHNR compliance without integrated accounting.
- There is **sensitivity and uncertainty around figures**, with a clear need for document-based verification and structured data collation before definitive numbers can be stated.

Direction for strengthening BHNR allocation planning (as implied in the interview)

The discussion concluded that the uncertainty around "how BHNR is allocated from bulk capacity" should be treated as a **structured investigation**, rather than expecting immediate answers from operational staff. The proposed direction—consistent with the interview—is to work with the district to:

- identify priority treatment plants and supply zones,
- verify **volumetric production and delivery** using metered information,
- map supply zones and populations,
- develop demand estimates (including BHNR minimums and realistic use profiles), and
- establish where systems are **under- or over-capacity** relative to BHNR requirements and actual service realities.

This provides a practical pathway to move from an implicit "capacity-and-operations" allocation approach toward a **transparent supply-demand accounting framework** that can support BHNR implementation and accountability at the level of specific supply zones and communities.

3.6 Practical actions to strengthen BHNR delivery through municipal planning and operations

1. Explicit BHNR accounting in WSDPs and municipal water balances (basic allocation volumes, service areas, and performance against minimum benchmarks).
2. Ring-fencing the basic allocation in operations: rules for restrictions, pressure management, and outage response that prioritise household minimum supply.
3. Strong NRW governance: measurement, targets, budgets and timelines aligned to the national standards' NRW and ILI expectations.

4. Use the standards' emergency supply rules to define and audit tanker plans, records, and quality monitoring during extended outages.

5. Publish simplified, local-language summaries of performance (continuity, quality advisories, NRW, outage response times) and enable accessible grievance channels.

Figure 14 Community standpipes in rural villages are frequently dray as they are not connected to a reliable or viable supply system

3.7 Municipal constraints and BHNR actions: evidence from the WRC (2025) assessment

The WRC (2025) review identifies financial stress, chronic under-maintenance and constrained procurement as central drivers of unreliability in rural WSAs. Large, complex regional systems prove brittle under weak procurement and delayed repair cycles, while losses and non-revenue water absorb resources that are supposed to secure minimum access. For WSAs/WSPs seeking to make the BHNR real, the assessment supports the following actions, which extend the practical measures in Section 3.2:

1. **Treat the BHNR as a protected first call in operations:** define explicit minimum-supply rules for distribution under intermittent supply, so that basic human needs volumes are the last to be interrupted and the first to be restored.

2. **Institutionalise routine (monthly) water balances as BHNR evidence:** system input, authorised consumption, losses, outages and emergency supply — making the minimum auditable rather than assumed.

3. **Link BHNR credibility to maintenance and repair performance:** mean time to repair (MTTR), spares availability and contractor turnaround time are rights-enabling indicators, not secondary engineering concerns.

4. **Reform supply-chain management for timeous repair** (a headline WRC recommendation): pre-approved supplier panels, standardised parts lists and specifications for common rural systems, and delegated rapid-procurement mechanisms within proper controls, so that repairs do not stall for months under MFMA processes.

5. **Favour modular, adaptive and decentralised technologies over large regional schemes,** especially for remote settlements not yet served — the WRC review argues

explicitly for a move away from expensive, complex solutions towards simpler, more affordable options, aligning with the MUS/service-ladder logic and supported self-supply models discussed in this report.

6. **Use partnerships for aspects of delivery where in-house capacity is stretched:** the review points to successful examples among the C2 districts of community-based organisations, water boards and public–private partnerships providing elements of the service.

7. **Adopt a realistic financing logic:** cost-reflective tariffs are unaffordable in remote rural contexts, communal areas are largely unbilled under Free Basic Water practice, and even doubling service income would leave most C2 districts short. BHNR delivery planning therefore requires a blended finance narrative — targeted operational support, transparent community contributions and fit-for-purpose grants — rather than assumed cost recovery.

Making the minimum visible: village-level metrics

Where reliability is weak and national datasets are incomplete — as with the withheld Census 2022 reliability results — community and operational evidence becomes essential to demonstrate whether the minimum right is being met. The WRC (2025) village assessments therefore strengthens the case for the village monitoring toolkit of Section 4.2, with emphasis on:

1. **Point-of-use measures:** litres per person per day actually received, days with supply, outage duration, and distance/time to collect — the lived equivalents of the allocation-side indicators used at catchment level.

2. **Coping costs and inequities:** purchases from vendors, storage investments and collection time — crucial evidence of unmet basic needs where official tables would otherwise show “0” or full coverage.

3. **Feeding monitoring into WSA reporting cycles and grievance mechanisms:** so that village evidence forms part of the rights-based accountability package and the catchment feedback loop.

3.8 BHNR Loss Factor Guide: Calculating the Total Loss Factor (f)

This section explains how to calculate the **Total Loss Factor (f)** used in **Basic Human Needs Reserve (BHNR)** assessments to convert a *theoretical minimum* water requirement (e.g., 25 litres/person/day) into a **realistic abstraction and supply volume**. In practice, not all water put into a rural water supply system reaches households as usable, safe water. Losses occur across the full delivery chain—from the source and bulk conveyance, through storage and distribution, to the point of use. The Total Loss Factor provides a transparent way to account for these losses so that BHNR estimates better reflect **actual system performance**.

The purpose of applying a loss factor is not to "inflate" the BHNR, but to **make the BHNR deliverable** under real operating conditions. Where systems are fragile, intermittently operated, poorly maintained, or affected by illegal connections and leakage, the gap between *what should be delivered* and *what is actually received* can be substantial. If losses are ignored, BHNR calculations can underestimate the water that must be abstracted, treated and conveyed—resulting in persistent shortfalls, unreliable services, and weak accountability.

The guide therefore sets out a practical method for calculating **f** as a composite of key loss components, typically including:

1. **Physical losses** (leakage from pipes, tanks, standpipes, and household connections)
2. **Operational losses** (overflows, poor pressure control, intermittent pumping, start-stop inefficiencies)
3. **Commercial/non-revenue losses** (illegal use, unmetered consumption, inaccurate or absent metering)
4. **Water quality and reject losses** (treatment backwash, flushing, or water discarded due to contamination risk)

By calculating and documenting **f**, implementers can (i) estimate the **required system input volume** to reliably meet BHNR at household level, (ii) identify the **largest drivers of loss**, and (iii) link BHNR delivery to **specific operational improvements**—such as leak repair, storage upgrades, pressure management, metering, and maintenance routines. Importantly, the Total Loss Factor also supports clearer **role allocation and accountability**, because it makes explicit whether shortfalls are driven by resource constraints, infrastructure condition, or operational management.

In the sections that follow, the guide defines the loss components, provides calculation steps, recommends default values where data are limited, and explains how to update **f** as monitoring improves—so that BHNR estimates can evolve from a conservative baseline to a robust, evidence-based operational requirement.

Core equations used in BHNR accounting

To avoid "paper BHNR" that disappears through losses or rounding, distinguish net (at point of use) from gross (at source).

Table 7: Core equations used in BHNR accounting

Definition	Equation
Net BHNR (annual volume)	*Net BHNR (m^3/a) = P × L × 365 ÷ 1000 where P = qualifying population (persons), L = litres/person/day (e.g., 25)*

Definition	Equation
Gross BHNR (loss-adjusted source requirement)	*Gross BHNR = Net BHNR ÷ (1 - f) where f = total loss factor (0-1), including NRW and conveyance/treatment losses*
Net BHNR (flow rate)	*Net BHNR (L/s) = (P × L) ÷ 86 400 Useful for operating rules and drought management*

You calculate f (the total loss factor) as the fraction of water put into the system (at source / treatment works outlet) that does not end up as delivered, usable water for households.

In other words:

$$\text{Gross BHNR} = \frac{\text{Net BHNR}}{1 - f}$$

f = (losses + non-household diversions that erode BHNR delivery) ÷ system input

(1 − f) is the delivery efficiency (the share that actually reaches households as intended)

Step 1: Decide the system boundary (be explicit)

Pick the boundary that matches what you mean by “Gross BHNR at source”. Common options:

Catchment abstraction to household delivery (broadest)

Treatment works outlet to household delivery (distribution-focused)

Bulk offtake to household delivery (municipal bulk + reticulation)

The *f* must be calculated within one defined boundary, otherwise it becomes meaningless.

Method A (recommended): Combine stage-specific losses (multiplicative)

If you can estimate losses by stage, calculate f from the product of efficiencies.

Let:

fc = conveyance/bulk transfer loss fraction

ft = treatment works loss fraction

fd = distribution loss fraction (physical NRW + apparent losses, etc.)

Stage efficiencies are:

$$\eta_c = 1 - f_c, \;\; \eta_t = 1 - f_t, \;\; \eta_d = 1 - f_d$$

Total delivery efficiency:

$$\eta_{\text{total}} = \eta_c \times \eta_t \times \eta_d$$

Total loss factor:

$$f = 1 - (1 - f_c)(1 - f_t)(1 - f_d)$$

Then:

$$\text{Gross BHNR} = \frac{\text{Net BHNR}}{1 - f}$$

Why this is better: it avoids double counting and reflects that losses occur sequentially.

Quick example

Assume:

> fc = 0.05 (5%) conveyance losses
>
> ft = 0.03 (3%) treatment losses
>
> fd = 0.35 (35%) distribution losses

Calculate efficiency and f:

$$\eta_{\text{total}} = (0.95)(0.97)(0.65) = 0.59975 \approx 0.60$$

$$f = 1 - 0.59975 = 0.40025 \approx 0.40$$

So only ~60% reaches households; gross BHNR must be:

$$\text{Gross BHNR} = \frac{\text{Net BHNR}}{0.60} \approx 1.67 \times \text{Net BHNR}$$

Method B: Use a single NRW value (when you only have high-level data)

If the only reliable figure you have is NRW (often reported as % of system input), you can approximate:

$$f \approx \text{NRW}$$

Then:

$$\text{Gross BHNR} = \frac{\text{Net BHNR}}{1 - f}$$

Caution: NRW alone usually reflects distribution/system losses and apparent losses; it may not include raw water conveyance losses, treatment losses, or operational releases, so it can understate f if your boundary is "source-to-tap".

Method C: Compute f directly from a water balance (best for audits)

If you have an IWA-style water balance (or monthly system input and billed/authorised consumption):

Let:

> SIV = System Input Volume (water entering the distribution system, typically at treatment outlet)
>
> Delivered (intended) = volume that actually reaches households for BHNR purposes (measured or estimated)

Then:

$$f = 1 - \frac{\text{Delivered (intended)}}{\text{SIV}}$$

If you cannot directly measure "delivered intended", you can use metered household consumption plus an estimate of unmetered household consumption, and then be explicit that you're calculating losses relative to household delivery, not total authorised consumption.

Important note for BHNR: don't mix "loss" with "diversion"

For BHNR ring-fencing, three different things can erode BHNR delivery:

- Technical losses (leaks, bursts, overflows)
- Administrative/apparent losses (meter under-registration, illegal connections)
- Diversion within the network (water going to non-household uses)

Only (1) and (2) are "losses" in the strict NRW sense. But (3) can still mean BHNR isn't achieved, so some projects define an "effective loss factor for BHNR delivery" that includes diversion risk.

If you do that, label it clearly, e.g.:

f NRW (technical + apparent losses)

f BHN-erosion (NRW + diversion away from BHNR delivery)

Practical tip: choosing *f* when data is weak

If you're doing scenario planning and only have rough NRW:

Best-case: f = 0.25 (25% losses) → efficiency 0.75

Typical stressed system: f = 0.40 (40% losses) → efficiency 0.60

High-loss system: f = 0.55 (55% losses) → efficiency 0.45

Then show gross BHNR multipliers:

$$\text{Multiplier} = \frac{1}{1-f}$$

$$f = 0.25 \Rightarrow \times 1.33;\ f = 0.40 \Rightarrow \times 1.67;\ f = 0.55 \Rightarrow \times 2.22$$

Use the following guide:

- Basic minimum service quantity: 6 kℓ/household/month (basic + interim)
- Flow rate: ≥ 10 ℓ/min
- Continuity: ≥ 358 days/year; no >48-hour interruption
- Emergency minimum during interruptions: ≥ 15 ℓ/person/day
- NRW/water loss performance expectations: NRW 20–30%, losses 10–20%, ILI 2–4; per capita usage benchmark 120–180 ℓ/c/d

- Monthly water balance + quarterly reporting requirements to the national regulatory system

3.9 Aligning villages with catchment allocations: an accountability chain

For BHNR to be credible, catchment allocations must connect to municipal plans and operational performance. A useful accountability chain is: catchment Reserve set-aside → municipal source allocation → loss-adjusted supply plan → distribution performance → household outcomes. Without this chain, reallocation debates can occur without any improvement in household access.

4. Village pathways: community and household practice

Village-level evidence makes the BHNR real. It shows whether minimum service outcomes are achieved and whether institutional responses are effective. It also reveals where catchment allocations and municipal plans fail to translate into reliable access because of infrastructure failures, operational choices, or inequitable distribution.

4.1 A practical village monitoring toolkit aligned to national standards

1. Community monitoring should be light-touch and consistent, using a small set of indicators that map onto enforceable benchmarks.
2. Continuity log: record days with supply; note any interruption longer than 48 hours; record whether emergency supply was provided and how.
3. Access and time: measure approximate distance to the nearest functional communal point (where relevant) and time spent collecting water.
4. Functionality register: track which standpipes/boreholes/storage tanks work, for how long, and how quickly repairs occur.
5. Simple flow proxy: use a bucket-and-timer test (where safe) to estimate litres per minute and compare to the 10 L/min benchmark.
6. Quality and risk: note taste/colour/odour changes; record boil-water advisories; record sources used during outages and whether they are protected.
7. Complaint tracking: record complaint dates, reference numbers, responses, and outcomes to build an evidence trail.

4.2 Community reporting and escalation pathways

1. Evidence must be actionable. A practical escalation chain is:
2. Report to the municipal consumer care facility and obtain a reference number.
3. Escalate unresolved issues through ward councillor/committee structures and municipal technical leadership.
4. Where failures are systemic, ensure evidence is tabled in IDP/WSDP review processes and district/provincial oversight engagements.
5. Where failures reflect resource constraints or upstream impacts, table evidence in CMA forums and CMS review processes.

4.3 Ensuring participation and non-discrimination in village partnerships

To generate a useful case study for this section work undertaken for the Water Research Commission entitled *Post project cross-sectional project implementation review: Operationalising Community-Led Multiple Use Water Services (MUS) in South Africa,* has been summarised to indicate what is possible and what is actually happening in existing multiple-use systems that are community-led.

The MUS programme focused on improving water access at household level through, rehabilitation of existing infrastructure, integration of multiple water sources, low-cost, community-driven upgrades and support for productive water use. Decision-making typically involves three key actors: Community users, implementing agents and higher-level government authorities.

Across all six communities, several common characteristics emerged:

- Households rely on multiple water sources as a coping strategy.
- Water is used for domestic, livestock, and productive purposes.
- Self-supply systems play a significant role alongside public infrastructure.
- Water sharing between households is widespread and often informal.

The six case-study communities represent diverse contexts in terms of water availability, infrastructure, and socio-economic conditions.

The six communities were selected in collaboration with the municipalities and the provincial and district officials of the Department of Water and Sanitation and the Limpopo Department of Agriculture and Rural Development. The villages represent a diversity in current service levels, investments in infrastructure for self-supply and existing public infrastructure, surface or groundwater resource availability, degrees of productive water users; number of people; etc.

The table below provides a summary of the MUS interventions in each of the 6 villages.

Table 8: MUS interventions in each of the 6 sites between 2017-2019 (TWS/IWMI)

Area	Type of system	Works	Comments
Sekhukhune			
Phiring 420HH	*Gov: Municipal borehole systems Dam and gravity pipeline for irrigation and other uses*	*Refurbish storage, repair reticulation Augmenting water supply to dam and repair leaks, extension to cattle dam 15 JoJos*	*Relocation of intake to dam - dam runs dry in winter Infrastructure is poorly maintained Reliability is inconsistent Livelihood potential remains underutilised*
GaMokgotho 850HH, 4 sections	*NGO: supported self-supply: gravity supply from springs, self-supply O&M:*	*Augmentation of supply, upgrades, repairs and extension of reticulation, 15 JoJos 2 drinking troughs*	*Lack of comm org for yard connections Breakdown of payment systems and governance structures Illegal household connections Inequities in water distribution Irrigation increased significantly, with notable gains in crop production and household income.*
GaMoela 118HH	*Gov: 2 Municipal borehole systems: Tawaneng/Letlabela and Mabusa / Moela borehole Communal handpumps Self-*	*New storage (4x1000l JoJos on stands) and reticulation Repairs 15 JoJos Drinking troughs*	*Need for booster pump for one borehole Many new taps Improved access and quality through MUS interventions Continued dependence on municipal support (electricity, maintenance) Persistent inequalities between sections*

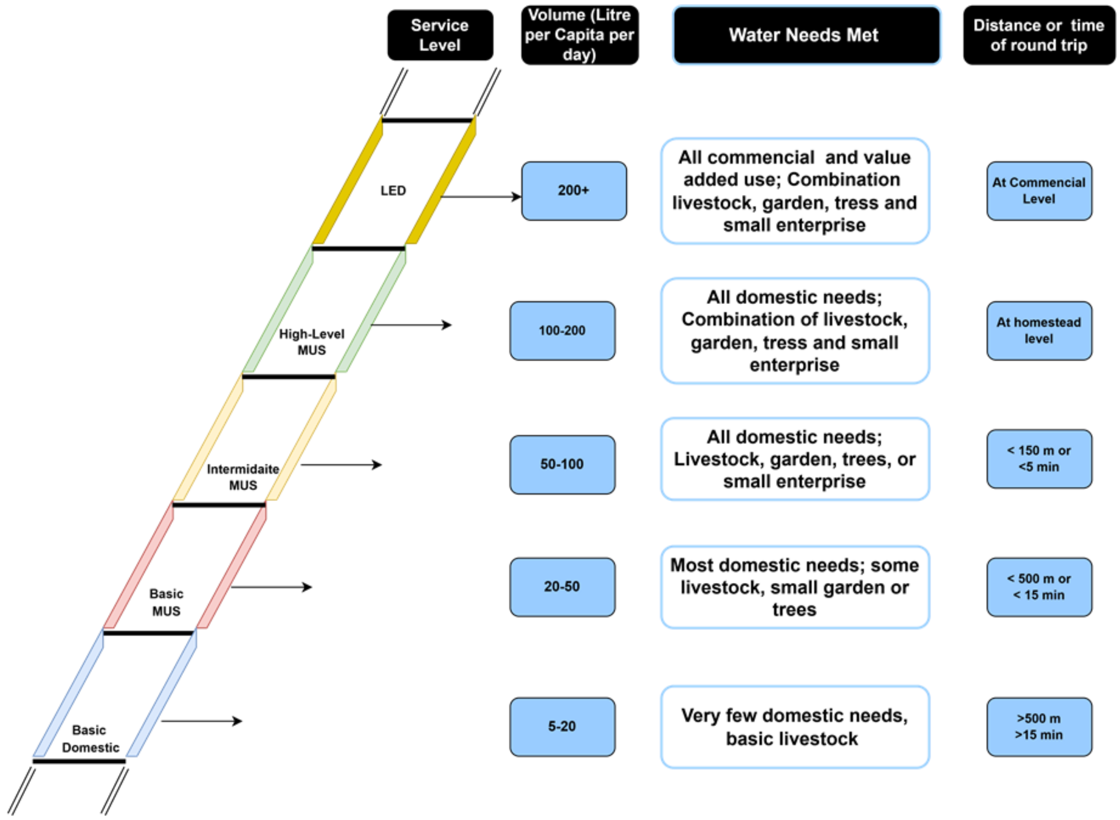
Area	Type of system	Works	Comments
	supply: hand-dug wells		
Vhembe			
Ha Gumbu 1652HH,3 sections	*Gov: Municipal borehole system Self-supply: HH boreholes with electrical pumps*	*Repair pumphouse, augmentation storage, extension of reticulation, Repair cattle trough 15 JoJos*	*Municipal borehole broken- only fixed after 2020- still to assess impact. Good markets. Strong household-level investment in water systems Weak communal infrastructure reliability Tensions around equity (e.g., JoJo tank allocation)*
Tshakhuma 2360HH, 9 sections (excluding Luvhalani and Tshitavhadulu)	*Gov: Municipal borehole Maswie Self-supply: 11 Piped gravity systems*	*Connecting to new storage (5x1000l JoJos and reticulated to HH taps - 220HH) Source development, augmentation storage, on new system 15 JoJos Drinking troughs*	*Protection of very distant springs High levels of community investment and innovation Significant livelihood benefits from improved water access Persistent inequalities linked to affordability*
Khalava 163HH	*Self-supply: Piped gravity system Communal hand pumps*	*Source development, new Storage Repairs 15 JoJos Drinking troughs*	*Lack of comm org for yard connections. Improved reliability and livelihoods in upgraded sections Continued disparities between sections Strong potential for scaling community-led approaches*

Village-level evidence makes the BHNR real. It shows whether minimum service outcomes are achieved and whether institutional responses are effective. It also reveals where catchment allocations and municipal plans fail to translate into reliable access because of infrastructure failures, operational choices, or inequitable distribution.

Within this context, MUS and supported self-supply systems also need to focus clearly on water availability, access, allocations and issues of equity.

4.4 MUS service ladder interpretation

The MUS 'service ladder' frames water service levels in terms of (i) quantity (liters per capita per day), (ii) distance/time to collect water, and (iii) the range of domestic and productive needs that can realistically be met.



Source: Smits, S.; van Koppen, B.; Moriarty, P. and Butterworth, J. 2010. Multiple-use services as an alternative to rural water supply services: A characterisation of the approach. Water Alternatives 3(1): 102-121

A simplified interpretation (based on the multiple-use ladder used in MUS literature) is summarized in the table below.

Table 9: MUS service ladder

Service level	Indicative quantity	Typical needs met	Distance/time (indicative)
Basic domestic	*5-20 l/person/day*	*Very few domestic needs; basic livestock (limited)*	*Often >500 m or >15 min*
Basic MUS	*20-50 l/person/day*	*Most domestic needs; some livestock; small garden/tree*	*Often <500 m or <15 min*
Intermediate MUS	*50-100 l/person/day*	*All domestic needs; livestock; garden/trees; small enterprise*	*Often <150 m or <5 min*
High-level MUS	*100-200 l/person/day*	*All domestic needs plus more reliable productive use at homestead*	*At homestead / yard connection*

With respect to the review of the 6 MUS systems, design choices about storage, reticulation density, and source reliability influence where communities sit on the service ladder. Design documentation indicates that upgraded system designs in some sections targeted substantially higher allocations than the Free Basic Water benchmark of 25 l/person/day, with indicative designs in the ~88–91 l/person/day range in some cases. Other contexts, such as spring-fed schemes planned for ~43 l/person/day, sit closer to the 'Basic MUS' band. These allocations are meaningful because they shape what communities can do (domestic convenience, gardens and livestock) and they affect institutional expectations and conflict risks if service levels are not transparently planned and governed.

The table below provides an indicative positioning of the six areas on the MUS service ladder. This is a qualitative interpretation, based on observed access arrangements (yard taps vs standpipes), reliability patterns and evidence of multiple-use activities. It should be updated if additional quantitative l/p/d measurements become available.

Table 10: MUS ladder levels for the 6 MUS systems assessed

Area	Indicative ladder level(s)	Notes (evidence basis)
Ga-Mokgotho	*Basic MUS to Intermediate MUS (varies by section)*	*Gravity-fed spring system with section valves and header tanks; multiple-use activities reported; some reticulation and storage constraints and vandalism issues.*
Ga-Moela	*Mixed: from Abandoned/Dysfunctional (collapsed sections) to Basic/Intermediate MUS (functional sections)*	*Portfolio includes reticulated boreholes functional sections with cost-sharing and improved reliability, and at least one collapse outlier (Pudi) with prolonged absence of supply.*
Ha-Gumbu	*Mixed: Basic domestic to Basic MUS*	*Unequal coverage from boreholes across sections; some households rely on hand pumps or long collection; infrastructure exposure and governance challenges reported.*
Khalavha	*Basic MUS to Intermediate MUS*	*Multiple sources (springs and municipal system) and productive water use reported; governance and infrastructure vary across blocks.*
Phiring	*Often Basic domestic to Basic MUS, with critical low institutional integration in some systems*	*Multiple sources, (springs, dam, municipal).Evidence of chronic infrastructure leakage and weak municipal support; some households rely on unsafe streams; multiple systems (boreholes, dam, Vrystad).*
Tshakhuma	*Intermediate MUS in best-performing sections; mixed across sub-sites*	*Strong community self-supply from springs 9and a borehole for one section) and governance in several sections; yard taps in some areas; constraints include pressure management, storage and uneven coverage.*

An analysis of household provisioning and allocations, in relation to the BHNR and the MUS ladder was undertaken. A summary table is provided overleaf outlining this analysis.

From the above table the allocations can be further summarized as shown in the table below.

Table 11: Water service levels by village

Per-capita supply (litres per person per day) and resulting service-level band for each village. Area rows show the area-average Lppd.

Area / Village	Lppd	BHN ≤25 l	Basic MUS >25-50 l	Intermediate MUS >50-100 l	High-level MUS >100-200 l
ga Moela	**36,2**				
Letlabela	47,6		✓		
Mabusa	33,8		✓		
Moela	33,6		✓		
Pudi	0,0	✓			
gaMokgotho	**31,0**				
Tawaneng	65,9			✓	
Lewhareng	36,4		✓		
Segabeng	14,3	✓			
Sethogeng	36,4		✓		
haGumbu	**18,1**				
Gumbu 1	20,3	✓			
Gumbu 2	20,0	✓			
Gumbu 3	14,0	✓			
Khalava	**104,2**				
Tondoni, Headertank 1 (highest)	50,0		✓		
Tondoni, Headertank 2	100,0			✓	
Tondoni, Headertank 3	166,7				✓
Tondoni, Headertank 4 (lowest)	100,0			✓	
Tshakhuma	**39,7**				
Dzananwa	28,0		✓		
Lukau	28,6		✓		
Maswie	60,0			✓	
Mathava	19,8	✓			
Motshindoni	25,9		✓		
Muhovhoya	17,1	✓			
Mulangaphuma 1	85,7			✓	
Mulangaphuma 2	42,9		✓		
Tshwishwiwiri	49,4		✓		
Phiring	**–**				
6 villages (equal amounts)	128,0				✓
Vrysad	0,0	✓			

Area / Village	Lppd	BHN ≤25 l	Basic MUS >25-50 l	Intermediate MUS >50-100 l	High-level MUS >100-200 l
Stocks & Stocks	-				
Totals: 6 areas, 27 villages	**46,9**	**30%**	**41%**	**19%**	**11%**

Band thresholds: BHN ≤25 · Basic MUS >25-50 · Intermediate MUS >50-100 · High-level MUS >100-200 l/p/d. Boundary values fall in the lower band (50,0 → Basic; 100,0 → Intermediate). Stocks & Stocks has no recorded Lppd and is unclassified.

This table serves to illustrate that the allocations within the villages for each area can differ substantially. Both gaMokghotho and Tshakhuma are examples of this where the different villages can fall anywhere between basic domestic supply and intermediate MUS levels. In Khalava different villages fall between Basic MUS, Intermediate MUS and high-level MUS levels.

As discussed thus far, the lower daily allocations for some of the villages, which fall below the BHNR are by and large institutional and technical, with malfunctioning boreholes and infrastructure, lack of adequate storage and leakages being the main stumbling blocks. Tshakhuma is the only area where some of the villages, by necessity, rely on springs that cannot sustain the number of households serviced.

The MUS systems in general for all 6 systems together provides around 47L per person per day – which is almost double the BHN allocated through institutional water service providers. This allows for household use primarily and provides a contribution to livelihoods but doesn't allow for any commercial production. All these systems allow for a high level of equity through internal village-level governance and also require financial contributions from the participating households, further allowing for a low but consistent level of operation and maintenance financing the villages.

In the two areas, Phiring and Khalava, which fall within the high-level MUS category, irrigation of larger fields and homestead plots is an option and commercial farming is undertaken in these areas by roughly 16% of the households. The same applies in haGumbu, except that irrigation is exclusively undertaken through the use of private boreholes in this case.

In summary the trends in terms of allocation and equity in these MUS systems are:

1. Those households with access to more storage (read more resources) have access to more water in all of the systems. In some cases, households with more water assist those who struggle with access freely. In other cases, the indigent households are expected to pay a small fee for this access.

2. Only higher volume systems make sense for household taps, and this appears to happen naturally, meaning that people have a good sense of how much water should be available before they implement the option of household taps. In most cases each household is expected to organize their own tap. This pattern also works against indigent households who cannot afford their own taps.

3. Even with an average of ~46L pppd, most villages restrict use of the system to household needs only and require that households find a different water source for irrigation.

4. Tap committees/ group for communal taps, as the lowest rung of allocation management in the villages works well for uncontested access in the villages. Sharing arrangements are negotiated between 8-10 people on average. Again, indigent households can be somewhat disadvantaged through these arrangements.

5. For villages in an area which have substantially lower access to water than their neighbouring villages, sharing arrangements are often put in place to assist them, but this only works when people feel they already have enough water and some to spare that can be shared. In areas like haGumbu the difference in access between the villages is substantial with high levels of conflict as water is not shared. In Tshakhuma the opposite is the case and water is shared quite freely and at no cost.

6. Even in villages with high levels of supply within the MUS ladder only around 16% of the households there have translated this provision into commercial production. This means that despite communities/ intense struggles to have access to more water, they cannot always capitalise on that resource, mostly due to financial and marketing constraints, exacerbated by climate change.

It is important to note that none of the systems are 100 percent equitable. They are, however, locally managed, and communities develop systems that are reasonably fair, reasonably non-conflictual and reasonably operational. This is a far cry from the largely unmanaged access created through the implementation of Government led water services. In addition, in all cases communities contribute financially and on an ongoing basis towards the management and upkeep of their MUS systems.

5. Cross-cutting synthesis: how to operationalise the BHNR

Across pathways, BHNR is most likely to be realised when the system makes three things explicit:

1. The minimum service outcome to be achieved (quantity, reliability, quality and emergency supply).
2. The loss-adjusted source requirement and the infrastructure/operations plan to deliver it.
3. The accountability chain: who must do what, by when, what evidence proves it, and what happens when it fails.

5.1 Priority action list

1. Adopt loss-adjusted BHNR calculations and state assumptions explicitly (NRW, conveyance losses, treatment losses).
2. Use reporting conventions that prevent BHNR requirements from rounding to zero (units, precision, percentage columns).
3. Scenario-plan population growth and service levels (25 L/person/day vs higher service targets) to make trade-offs explicit early.
4. Treat NRW reduction as a core BHNR delivery lever; couple reallocation debates with efficiency and operational improvement commitments.
5. Strengthen ring-fencing of the basic allocation within municipal systems through accounting, metering, and operational rules.
6. Institutionalise a catchment-to-village evidence pathway so that community service performance data can inform catchment strategies, drought response and allocation decisions.
7. Publish accessible performance information (service continuity, water quality compliance, NRW) and maintain an actionable grievance mechanism.

6. A proposed BHNR Delivery plan

Table 12: Frameworks and concepts underpinning the BHNR delivery plan

Framework / concept	How it is used in this delivery plan
Reserve (BHNR + Ecological Reserve)	*Highest-priority allocation. Sets a non-negotiable lifeline standard for domestic use while protecting aquatic ecosystems.*
RIDe (Resources-Infrastructure-Demands-entitlements)	*A practical way to diagnose whether problems originate in resource availability, infrastructure performance, demand patterns or institutional entitlements and rules.*

Framework / concept	How it is used in this delivery plan
MUS and SWELL	*Participatory methods to elicit real household uses and seasonality; design services that protect BHNR while enabling livelihoods (gardens, livestock and micro-enterprises).*
IWRM (catchment and sub-catchment)	*Operating rules and drought triggers align water allocation and restrictions with service delivery realities.*
Water Safety Planning	*Risk-based barriers from source to point-of-use, supported by routine monitoring and corrective actions.*

6.1 Objectives, results and success criteria

The objectives below define what the delivery plan must achieve and how success will be measured.

Table 13: Delivery plan objectives, key results and success criteria

Objective	Description	Key results / indicators
Objective 1	*Guarantee BHNR service standard in target rural wards.*	*Uptime >=95%; access distance <=200 m; >=95% E. coli pass; >=25 L/person/day for >=90% households.*
Objective 2	*Institutionalise O&M and governance.*	*Framework repair contracts with 48h/5-day SLAs; ward-level spares caches; signed scheme MoUs (WSA-community).*
Objective 3	*Integrate MUS without compromising BHNR.*	*Lifeline (free/near-free) plus post-BHNR volumetric tariff; MUS add-ons at priority sites.*
Objective 4	*Align catchment allocation and municipal planning.*	*Reserve-aware operating rules; WSDPs updated using RIDe outputs.*
Objective 5	*Establish performance monitoring and learning.*	*Village scorecards; municipal dashboard; annual learning review.*

6.2 Implementation pathway

This **BHNR Delivery Plan** sets out a practical “catchment-to-village” pathway for turning the **Basic Human Needs Reserve (BHNR)** from a legal entitlement into **reliable water at household level** in rural South Africa (e.g., Giyani/Bushbuckridge).

The Plan starts from a simple problem: **BHNR is a legal priority, but many rural systems remain unreliable**—with frequent breakdowns, water quality risks, and access distances that exceed the intended standard. It responds by integrating **water resource allocation (“Reserve first”)**, **municipal service delivery**, and **village-level functionality and accountability** in one delivery chain, so that the BHNR is not only set aside on paper, but can be demonstrated through service outcomes.

This Delivery Plan is anchored in an integrated set of frameworks—**Reserve (BHNR + Ecological Reserve)**, **RIDe (Resources-Infrastructure-Demands-entitlements)**, and **SWELL/MUS**—and it translates them into implementable actions, budgets, roles and performance measures. The goal is to **guarantee ≥25 litres/person/day**, delivered with **safe quality**, through service standards that are measurable (e.g., **≥95% uptime, ≤200 m access, and strong microbiological compliance**) while protecting the Ecological Reserve and explicitly managing post-BHNR **multiple-use water** for livelihoods.

The Plan is structured around four connected levels of implementation:

1. **Catchment & sub-catchment:** a resources audit, "Reserve-aware" operating rules, and drought protocols.
2. **Municipal/WSA level:** service standards, asset registers, O&M systems (including **repair SLAs and spares caches**), financing, and dashboards.
3. **Scheme level:** standardised technical packages, Water Safety Plans, appropriate monitoring, and security.
4. **Village level:** SWELL/MUS appraisals, simple MoUs on roles and entitlements, community micro-financing (e.g., VWISA), and public scorecards.

Finally, the Delivery Plan provides a phased work programme—**0-90 days (quick wins), 3-12 months (lock in reliability), 12-24 months (scale and resilience)**—alongside governance arrangements (DWS/CMA, WSA/municipality, VWISA/water committees, and partners), financing principles, risk mitigation, and an MEL framework to make BHNR delivery **visible, auditable, and improvable over time**.

Implementation is organised as four linked workstreams that connect catchment allocation to village-level delivery.

6.3 Catchment and sub-catchment (DWS/CMA, WSA & stakeholders)

C1. Water resources audit (RIDe-R): Source registry (dams, rivers, aquifers), dry-season safe yields, quality risks; drought bands (normal/dry/very dry).

C2. Reserve-aware operating rules: BHNR and ecological Reserve locked in; abstraction caps; seasonal triggers; emergency tanker protocols.

C3. Priority node mapping: Identify villages/schemes at risk (<1.2x BHNR in dry season); assign augmentation options (rehab, drill, interties).

Outputs

1. Catchment water balance by node.
2. Reserve-first rule set.
3. Drought protocol.

6.4 Municipal/WSA level (WSDP and service delivery)

M1. Service standard: Define lifeline BHNR: >=25 L/person/day, <=200 m, >=95% uptime, and quality criteria.

M2. Asset and functionality register (RIDe-I): Pumps, storage, pipes, energy, treatment, telemetry, spares status.

M3. O&M system: Framework contracts with SLAs (48h minor / 5-day major); ward spares caches (standardised parts); monthly preventative maintenance checklists; quarterly services; annual yield tests.

M4. Financing: Ring-fenced O&M from equitable share plus conditional grant top-ups; performance-based disbursements tied to uptime and quality KPIs.

M5. Monitoring: Municipal dashboard (uptime, repairs, quality, queue times).

Outputs

1. Updated WSDP annexures.
2. O&M contracts and budget lines.
3. Dashboard live.

6.5 Scheme level (bulk or local)

S1. Standard technology packages: Solar-powered borehole to elevated storage to gravity standpipes; simple chlorination; roughing/cartridge filters if needed; two or three standard pump/controller models for scale.

S2. Water Safety Plan (WSP): Hazards, barriers, critical limits and response actions.

S3. Telemetry-lite: Run-time counters or GSM loggers (where feasible).

S4. Security: Anti-theft cages, lighting, community watch rota.

Outputs

1. Scheme designs and bills of quantities (BoQs).
2. Water Safety Plans.
3. Security plan.

6.6 Village/community level (SWELL/MUS + governance)

V1. SWELL/MUS appraisal (RIDe-D): Household sizes, uses (domestic, gardens, livestock), seasonality, peak hours; identify vulnerable users.

V2. Governance and entitlements (RIDe-e): One-page MoU (WSA-VWISA/Committee): roles, hours, hygiene, fee rules. BHNR lifeline free/near-free; post-BHNR volumetric tariff for MUS.

V3. O&M micro-financing: VWISA petty cash for small repairs and chlorination; transparent ledger; public noticeboard.

V4. Scorecards and accountability: Monthly display of uptime, repairs, quality, funds in/out.

V5. Inclusion and safety: Women/youth representation; safe access; disability-friendly tap design.

Outputs

1. SWELL report.
2. MoU and tariff notice.

3. Village scorecards.

7. Conceptual and practical actions to be considered

1) First fix the conceptual mismatch: "BHNR at source" ≠ "water at the tap"

A recurring problem (and one consistently pointed at) is that BHNR is calculated as if there are no losses between the source and households. But the rights-based work is blunt: if we assume no externalities/losses, the BHNR "as currently set" becomes indefensible in real systems.

WHiRL's policy-facing synthesis made the same point in simpler terms: as much as twice the volume delivered "at the tap" can be needed at the source once losses are acknowledged.

Practical implication: when catchments "set aside" BHNR on paper, it can still fail people unless we also:

- translate it into gross source requirements (loss-adjusted),
- link it to infrastructure deliverability, and
- add controls so BHNR water isn't simply absorbed by NRW or higher-paying uses.

2) Make BHNR defensible in a water balance: calculate it as a loss-adjusted source requirement

A simple way to remove the "false zero" problem and make the number meaningful:

A. Start with BHNR (legal minimum)

- **Net BHNR = P × 25 L/p/d**

B. Convert to a gross source requirement

- **Gross BHNR = Net BHNR ÷ (1 − loss factor)**

Where "loss factor" should be explicit and separated, e.g.:

- bulk conveyance losses,
- treatment works losses,
- distribution NRW (physical + apparent).

This aligns with the rights-based argument that the BHNR must be high enough to achieve the minimum content at the tap, not just "exist" hydrologically.

3) Keep 25 l/p/d (BHNR) and 100–110 l/p/d (service level) in different policy boxes

Your scenario concern is real: moving from 25 →100/110 l/p/d can push balances into deficit and trigger reallocation fights.

The clean way to present this (and avoid legal/policy confusion) is:

- BHNR = the priority Reserve minimum (what must be protected first), and
- 110 l/p/d = the planned service level target (a municipal planning scenario, not necessarily a "Reserve" requirement unless policy explicitly changes).

Then run both as scenarios, but don't let the 110 number silently replace BHNR in Reserve tables.

4) Stop "rounding BHNR to zero": change the reporting units and conventions

This is a presentation problem that creates a governance problem.

Use one (or more) of these conventions so BHNR never disappears in tables:

1. Report BHNR in smaller units for balance tables:

   Ml/a (not Mm^3/a), or m^3/d, or L/s (often best for operating rules).

2. Increase decimal precision for priority items (e.g., 3–4 decimals for Mm^3/a only for BHNR/ER).

3. Add a second column showing BHNR as a % of available water (e.g., % of NMAR/assured yield).

4. Use scientific notation for tiny volumes instead of rounding.

That way, BHNR stays visible even when agriculture/mining dominate the totals—and you avoid the optics of "0" that communities correctly read as erasure.

5) Population growth and 10–25 year projections: use a range, not a single escalator

The most defensible approach for CMA/catchment planning is scenario-based projections:

- Base case: Stats SA / official projection baseline (district/municipal) where available.
- Low / high growth sensitivity: alternative growth rates to show uncertainty.
- Service-improvement trajectory: the "dependent population" may decrease if supply improves—unless system failure/NRW keeps dependence high.

DWS BHNR reporting practice is also moving toward separating BHNR reliant on surface vs groundwater and anchoring the assessment year explicitly (e.g., "year 2021 was determined").

6) The ecological reserve vs BHNR: don't let the trade-off stay implicit

The rights-based analysis you have flags the tension directly and (controversially) notes that BHNR is closer to a core obligation, while ER is often treated as progressive—yet long-term water security depends on ER integrity.

When 110 l/p/d scenarios create deficits, the responsible order of response is:

1. Reduce losses / NRW first (otherwise you're reallocating water just to leak it),
2. Demand management / restrictions across sectors,
3. Reoperation / augmentation where feasible,
4. Only then explicit reallocation debates (with transparent socioeconomic trade-offs).

7) Ring-fencing BHNR in municipal supply: treat it like a protected "first call" within the system

Even if a CMA "allocates" BHNR, municipal realities can redirect it (NRW, industry, commercial use).

A workable "ring-fence" package typically includes:

1. Separate accounting for BHNR volumes in WSDP / water balance reporting (not just total municipal allocation).
2. Metering + auditing at key nodes (bulk offtakes, treatment works inlet/outlet, DMA-level where possible).
3. Explicit licence / authorisation conditions (where applicable) that require reporting of BHNR performance indicators.
4. NRW performance targets linked to compliance and funding conditions.

South Africa's revised compulsory national water and sanitation services standards (gazetted 6 June 2025) are explicitly framed as enforceable minimum requirements with strengthened monitoring/accountability expectations—use these as the "regulatory hook" for ring-fencing and reporting.

8) The "new norms and standards" – a pathway to implementation

The standards were prescribed by the Minister of Water and Sanitation (Pemmy C.P. Majodina, MP) in terms of section 9(1) of the Water Services Act, 1997 (Act No. 108 of 1997), signed on 16 April 2025 and published on 6 June 2025 as Government Notice No. 6292 in Government Gazette No. 52814, under the short title "Compulsory National Water and Sanitation Services Standards, 2024". They repeal the 2001 compulsory national standards (General Notice Regulation 509 of 8 June 2001) and bind all Water Services Institutions.

The gazetted provisions that bear most directly on points 1–7 above are:

1. **Basic water supply (Regulation 2):** a minimum of 6 kℓ of drinking water per household per month at a flow rate of not less than 10 L/min, available at least 358 days per year, with no interruption longer than 48 consecutive hours, and compliant with SANS 241; an access or delivery point at least at the yard edge (user connection point) of existing settlements within two years of promulgation; and the initial 6 kℓ at no cost to indigent households. This is the regulatory floor that point 3 argues must be kept in a separate policy box from the 25 ℓ/p/d BHNR at source.
2. **Interim water services (Regulation 3):** a communal standpipe within 200 m of the furthest household, provided within 90 days of a new informal settlement being realised, with the same quantity, flow-rate, continuity and quality minimums as basic supply; reliance on water tankers may not exceed 12 consecutive months and full delivery records must be kept.

3. **Interruption of supply (Regulation 4):** where supply is interrupted for more than 48 hours, consumers must have access to at least 15 litres of drinking water per person per day at strategically determined, safe and relatively convenient delivery points — regulatory backing for the “protected first call” logic proposed in point 7.
4. **WCWDM performance (Regulation 16):** leaks repaired within 48 hours of becoming aware; 95% of detected pipe bursts isolated within 4 hours; reticulation pressures managed between 20 m and 90 m; and key performance indicators of non-revenue water 20–30%, water losses 10–20%, Infrastructure Leakage Index 2–4 and per-capita usage 120–180 ℓ/c/d — the authoritative loss parameters that point 2’s loss-adjusted source requirement should draw from.
5. **Water and wastewater balances (Regulation 17):** a monthly water balance measuring daily abstraction and treatment and the quantity supplied to each supply zone, determined per the IWA water-balance guideline, with results reported quarterly to the Department’s national regulatory information management system — the natural reporting vehicle for point 4’s non-vanishing units (Mℓ/a, m^3/d or L/s) and precision conventions.
6. **Audit, compliance and offences (Regulations 26–28):** an annual water services audit within four months of each municipal financial year-end, quantifying services, service levels as percentages of households, water-balance results, losses/NRW and O&M expenditure against budget; progressive-compliance plans submitted on IRIS within six months where immediate compliance is not achievable; and offences for contravening the prohibitions, failing to issue drinking-water advisory notices or failing to provide the required plans — machinery that converts the actions in points 1–7 from advocacy into monitorable compliance duties.

The defensible citation for all norms-and-standards claims in this report is the gazette itself (Department of Water and Sanitation, 2025), read together with the extraction and compliance analysis in Sections 1.3.1–1.3.2.

Concluding remarks

These case studies set out to answer a simple question: *if the Basic Human Needs Reserve is the highest-priority allocation in South African water law, why does it so often fail to arrive at the tap in rural municipalities?* The answer that emerges from the Sand River catchment assessment, and from the Bushbuckridge and Mopani District experiences, is that the BHNR fails not because the water is absent or the law is silent, but because nothing currently connects the allocation on paper to the service on the ground. The Reserve can be fully "present" in a catchment water balance while remaining absent in practice — lost to non-functional infrastructure, unrecorded losses, reporting conventions that round the basic requirement to zero, and an accountability chain in which no single institution can be shown to have failed.

The RIDe lens proved central to making this visible. By separating Resources, Infrastructure, Demands and entitlements, the assessment showed that the binding constraint on the BHNR is rarely hydrological. In the Sand River catchment, the decisive gaps lie in the middle of the chain: infrastructure whose downtime is neither measured nor consequential, demand projections that conflate the 25 ℓ/p/d lifeline with much higher service-level aspirations, and entitlements that assign the right to water without assigning the duty — and the evidence trail — to deliver it. This diagnosis matters because each failure type calls for a different remedy, and because treating the BHNR as a deliverability problem rather than an allocation problem redirects attention from reallocation debates towards operations, maintenance, loss reduction and verification.

The study's second contribution is the set of action pathways at national, catchment, municipal and village levels, which together constitute the beginnings of the integrated framework whose absence motivated this work. The gazetting of the compulsory National Water and Sanitation Services Standards in June 2025 has materially strengthened the national pathway: minimum quantities, flow rates, continuity limits, loss parameters and audit obligations now exist as binding, monitorable duties rather than policy aspirations. The task ahead is to make the other levels answerable to that floor — catchment operating rules and water balances that carry the BHNR as a loss-adjusted, ring-fenced first call; municipal Water Services Development Plans that translate the allocation into scheme-level performance commitments; and village-level monitoring that generates the evidence on which the whole accountability chain depends. The catchment-to-village pathway is deliberately bidirectional: allocations and rules flow downward, but the proof of realisation flows upward, from household experience through village scorecards to catchment and national reporting.

Several cautions temper this. Progressive realisation cannot become an indefinite deferral; the two-year and ninety-day timeframes in the 2025 Standards now give that principle enforceable edges. The trade-off between the BHNR and the Ecological Reserve, and between lifeline supply and multiple-use livelihoods, must be made explicit rather than resolved by default in times of shortage. The institutional effort of the proposed delivery plan is real: framework O&M contracts, community-level governance arrangements such as VWISAs, and routine water balancing all demand capacity that many rural Water Services Authorities do not yet have. The delivery plan in Section 6 is therefore best read as a sequenced pilot — proving the model in the Lowveld Rivers region before wider replication through the CCC — rather than a blueprint to be imposed at scale.

The overall conclusion is nonetheless one of practical optimism. The legal architecture for the BHNR is now substantially complete; the methods for diagnosing where it breaks down exist and have been demonstrated; and the 2025 Standards supply the compliance machinery that was previously missing. What remains is the disciplined work of implementation: calculating the BHNR as a loss-adjusted source requirement, keeping it visible in every water balance and report, contracting for reliability, and ensuring that the people the Reserve exists to serve are the ones who verify whether it has been delivered. If the pathways proposed here are followed, the phrase "from the catchment to the village" ceases to be a metaphor and becomes an auditable chain of custody for the most basic entitlement in South African water law

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Table 14: Detailed service assessments for selected villages in Limpopo Province

Area	Village	Source	Source yield (based on time to fill header tanks (L/s)	No of HH	storage (Litres)	time to fill (hrs)	HH allocation/week	HH allocation/day	Ave HH size	Ave Lpppd available	Ave Lpppd for those with 4 x210l	Lpppd for those with JoJos (	Comments
ga Moela	Letlabela	Borehole 1	0,3	23	3000 0	4	1666,7	238,1	5	**47,6**	25,0	164,3	2 sections - pumped 2x/week per section)- Wednesday and Saturday people have access.18hh as the 5 with JoJos receive separately
	Tawaneng	Borehole 1	0,3	34	8000 0	36	2307,7	329,7	5	**65,9**	25,0	285,7	10000estimate for JoJos every week . 30 000L x 2 as it is filled 2x/week, divided by the rest of the HH
	Mabusa	Borehole 2	1,0	23	30 000	8	1184,2	169,2	5	**33,8**	25,0	178,6	7500l goes directly to JoJos. The rest is shared by 19 people
	Moela	Borehole 2	0,7	17	20 000	3	1176,5	168,1	5	**33,6**	25,0	142,9	This is only half now as one tank is not operational
	Pudi	Borehole 3	0	16						**0**			Borehole is not functioning
gaMokgoth o	All	2xSprings	5,6	870	160 000	8	183,9	183,9	5	**36,8**			Here people also have pipes that water their trees, in addition to their JoJo tanks and drums - organised at tap level. Some households receive a lot more water, but it is difficult to quantify. Note: system operates every day of the week - so daily allocations aren't divided by 7.
	Sethogeng		1l/s and 4l/s	148	6200 0		418,9	182,1	5	**36,4**			Supplies Sethogeng and Nkoting, including 4 new HH. 2 valves, so each of three sections gets water very 3rd day.
	Segabeng			96	2400 0		250,0	71,4	5	**14,3**			

	Lewhareng			148	62000		418,9	182,1	5	**36,4**			the central line supplies Lewhareng, then Mamphawu and the bottom end of Sethogeng with roughly 20 new stands. The latter aren't supplied with water. So the rest of the 160KL allocation. However other supplies as well like TA, schools, etc - so remove 12KL.
Tshakhuma	Mulangaphuma 1	Spring	2,5	105	45000	1	428,6	428,6	5	**85,7**		500,0	HH access depends on individual stroage- these are filled every day. As the HH JoJo tanks are filled overnight- actual allocation/hh depends on the actaul amount of storge and can be as high as 2500-7000L/day
	Mulangaphuma 2	Spring	1,64	121	30000	?	1500,0	214,3	5	**42,9**		214,3	Actual HH access depends on individual storage- these are filled once a week. 75% of participants have JoJo tanks.As spring in Mulangaphuma1 is presently twic as strong as the origianl yield measured, the same has been estimated for this spring. 6 sections ~20hh per section..
	Lukau	Spring	3,25	210	30000		1000,0	142,9	5	**28,6**		142,9	In winter flow rate of the spring drops substantially (then only 3 x5000L tanks can be kept full. In summer there is substntial excess capacity, which is just left to overflow - as these is not extra storage.In this village, there are very few storage tanks. People just fill containers from their taps. Only 1 JoJo tanks (1000-5000L) per household is allowed
	Tshwishwiwiri	Spring	1,38	162	20000		1728,4	246,9	5	**49,4**			Leaks in header tanks cause a substantial reduction in potential of this system. In this village, there are very few stroage tanks. People just fill containers from their taps.

	Muhovhoya	Spring	0,6	350	3000 0		600,0	85,7	5	**17,1**		40	11 streets, (3 sections). Also get Municipal water Thursday through Sunday at household level taps. MUS system runs continuously. … If people use for only 3 days of the week - allocation is around 40lppd
	Maswie	Borehole	2,45	200	60 000	5	600,0	300,0	5	**60,0**		210	Section A receives water 2x/ week. Section B has 15 000L provided to people from that area and others 9mathava…) from 4 taps at the tanks - filled Wed, Saturday - 1hr. A lot of water for this system. No restrictions so JoJos can be filled2x/week. Around 60% of HH in the area have JoJotanks, directly connected to the system.
	Dzananwa	Spring/ stream	0,23	200	2000 0	5	500	140,0	5	**28,0**		50,7	Area divided into 5 streets - 1 day per street. Water every five days. Can fill a JoJo tank of 2500 (every 10 days). A private service provider cam in in 2024. He has tapped directly into the stream and thus provides continious flow for people participating - providing alot more water to them ,but at the cost of the ecological reserve. Flow from tanks continuous. 1 Day per streetx5 streets. Water finishes within 4-5hrs. 8,5% have boreholes and 5% have JoJotanks- mostly drums and containers
	Mathava	Spring/ stream	0,12	101	2000 0	5	990,1	99,0	5	**19,8**		71,4	3HH (3%) with 5000l JoJos and 13 HH (13%) with 2500L. Only 1 borehole0 at Musanda's house. JoJos filled 1x/week. Only get water once every two weeks. Lower section of village ~40HH have government tap water.

	Motshindoni	Spring/ stream	0,12	232	30000	5	354,6	129,3	5	**25,9**		71,4	Source shared with Mathava - assume same strength. Assume same strength of spring as Mathava (as it is shared). Also assume 5 streets.
Khalava	Tondoni, Headertank 1- higest	Spring/stream	0,69	40	10000	3	750	250	5	**50**	126		Allocations for this section includes a school and two churches as well, Water is provided every 2nd day, 3x/week. Average allocation pppd used to be 126L (3x210L drums/household). This has now reduced to 25L only . This water is for household use. Water for farming and other uses is provided through the old self-supply system and most people have JoJos tanks that they fill for this purpose.
Khalava	Tondoni, Headertank 2	Spring/stream	0,69	20	10000	3	1500	500	5	**100**	126		
	Tondoni, Headertank 3	Spring/stream	0,69	12	10000	3	2500	833,3	5	**166,7**	126	166,7	12 households have JoJo tanks connected to this system and gardens. Animal trough in this section is not used and has been disconnected.
	Tondoni, Headertank 4-lowest	Spring/stream	0,69	20	10000	3	1500	500	5	**100,0**	126		This tank fills up first ,as it is the lowest and water for hthis section is the most reliable. Sometimes the higher header tanks do not fill up
haGumbu	Gumbu 1	Borehole	4,16	296	30000	4	710	101,4	5	**20,3**			Municipal borehole upgraded with electrical pump, with storage and reticulation added by the MUS system. 130 private boreholes

	Gumbu 2	Borehole	1,38	100	10000	3	300	100	5	**20,0**			Municipal borehole upgraded with electrical pump, with storage and reticulation added by the MUS system. Only 1 of the 3x 10 000L header tnaks is functional
	Gumbu 3	Borehole	2	100	0	0	0	70	5	**14,0**			Municipal borehole upgraded with electrical pump, with storage and reticulation added by the MUS system. The borehole is not functional. 12 Household boreholes. There is no operator in Gumbu 3. There is an old borehole with a handpump in use - both for people and providing water for livestock.
Phiring	Mohlwatsengwane, Tshangane, Legwareng and Khalanyoni and section 2: Phiring and Khalanyoni	Borehole	8,6	420	3	92448	3080	440	5	**128,0**	88	88	Household taps and 2500l JoJo tanks for ~20% . Household taps for Sotcks&Stocks dam – only those who could afford (add 40lppd as estimate
	Stocks&Stocks: irrigation	Dam	174	67				45 000					67 irrigators, 250hain total – not all used(roughly 50%) 30 000- 60 000L /ha/day